Chapter 22

Area of a Trapezium and a Polygon

Class - 8 Concise Mathematics Selina

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CONTENTS

Exercise 22(A)

Question 1(i)

Two sides of a triangle are 12 cm and 15 cm. If the height of the triangle corresponding to 12 cm side is 10 cm, then the height of the triangle corresponding to 15 cm side is:

1. 4 cm
2. 12 cm

3. 8 cm

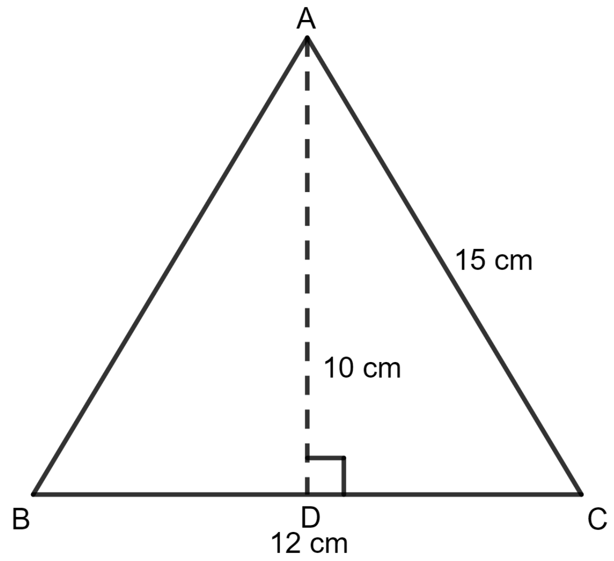
4. 16 cm

Answer

Given:

Sides of the triangle = 12 cm, 15 cm

Height of the triangle = 10 cm



As we know, the area of the triangle = $\frac{1}{2}$ x side x height

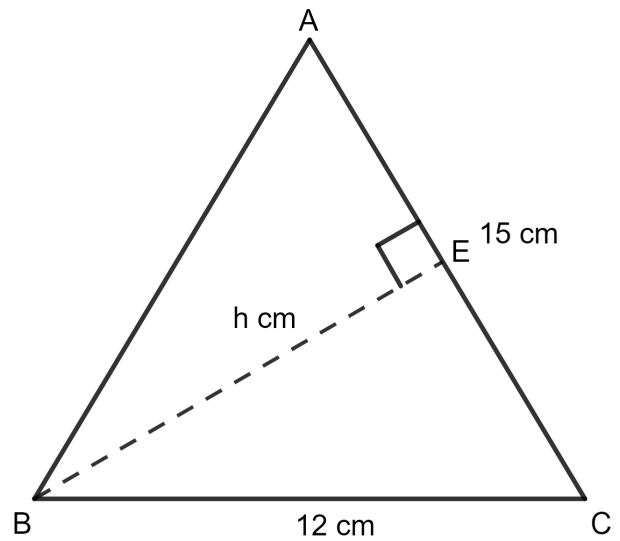
= $\frac{1}{2}$ x 12 x 10 cm^2

= $\frac{1}{2}$ x 120 cm^2

= 60 cm^2

When, side of the triangle = 15 cm

Let the height of the triangle = h cm



As we know, the area of the triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 15 x h ^{2}

[∵ Both areas remain same since they represent the area of the same triangle.]

⇒ $\frac{1}{2}$ x 15 x h = 60

⇒ 15 x h = 2 x 60

⇒ 15 x h = 120

⇒ h = $\dfrac{120}{15}$

⇒ h = 8 cm

Hence, the height of the triangle corresponding to 15 cm side is 8 cm.

Hence, option 3 is the correct option.

Question 1(ii)

The base of a right-angled triangle is 8 cm and its hypotenuse is 10 cm. The area of the triangle is:

1. 80 cm^2
2. 40 cm^2
3. 48 cm^2
4. 24 cm^2

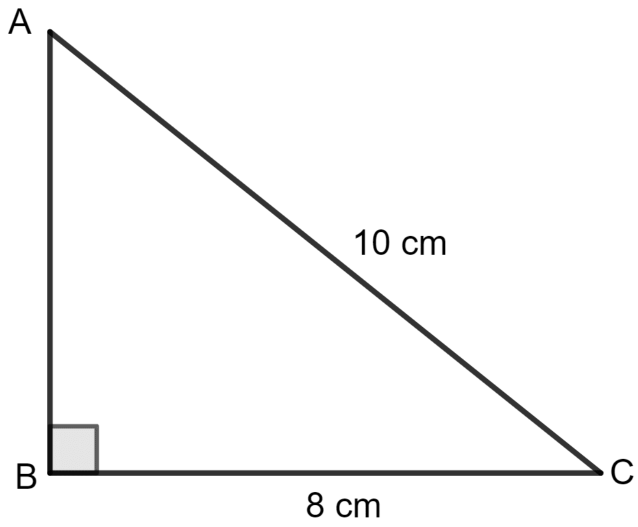
Answer

Given:

Base of the triangle = 8 cm

Hypotenuse of the triangle = 10 cm

Let `h` be the height of the triangle.



In a right-angled triangle, using the Pythagorean theorem,

$\text{Base}^2 + \text{Height}^2 = \text{Hypotenuse}^2$

$\Rightarrow (8)^2 + h^2 = (10)^2$

$\Rightarrow 64 + h^2 = 100$

$\Rightarrow h^2 = 100 - 64$

$\Rightarrow h^2 = 36$

$\Rightarrow h = \sqrt{36}$

$\Rightarrow h = 6$ cm

As we know, the area of the triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 8 x 6 cm^2

= $\dfrac{1}{2}$ x 48 cm^2

= 24 cm^2

Hence, option 4 is the correct option.

Question 1(iii)

The area of a triangle with sides 4 cm, 3 cm and 5 cm is:

1. 12 cm^2
2. 15 cm^2
3. 10 cm^2
4. 6 cm^2

Answer

Let a = 4 cm, b = 3 cm and c = 5 cm.

$\because s = \dfrac{a + b + c}{2}$

$= \dfrac{4 + 3 + 5}{2}$

$= \dfrac{12}{2}$

$= 6$

∵ Area of triangle = $\sqrt{s(s - a)(s - b)(s - c)}$

= $\sqrt{6(6-4)(6-3)(6-5)}$ cm^2

= $\sqrt{6 \times 2 \times 3 \times 1}$ cm^2

= $\sqrt{36}$ cm^2

= 6 cm^2

Hence, option 4 is the correct option.

Question 1(iv)

The altitude of an isosceles triangle is 4 cm and its base is 2 cm more than its height. The area of the triangle is :

1. 24 cm^2
2. 12 cm^2
3. 16 cm^2
4. 8 cm^2

Answer

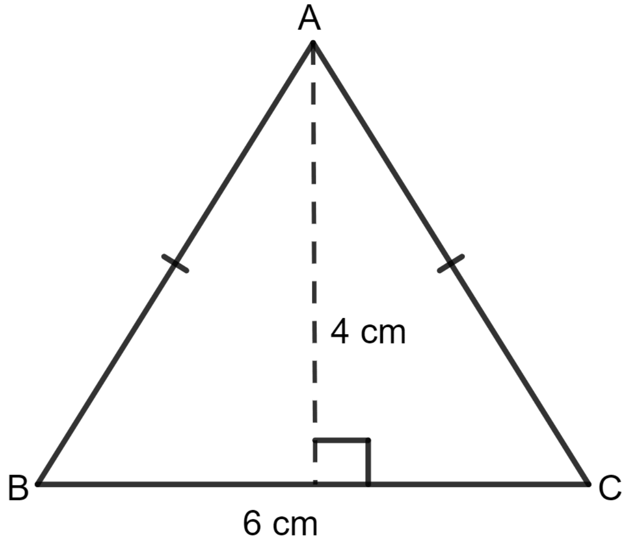
Given:

Altitude of the triangle = 4 cm

Base of the triangle = 2 cm more than its height

= 2 + 4 cm

= 6 cm



∵ As we know, the area of the triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 6 x 4 cm^2

= $\frac{1}{2}$ x 24 cm^2

= 12 cm^2

Hence, option 2 is the correct option.

Question 1(v)

Area of an equilateral triangle with each side 6 cm is:

1. $36\sqrt{3}$ cm^2

2. $9\sqrt{3}$ cm^2

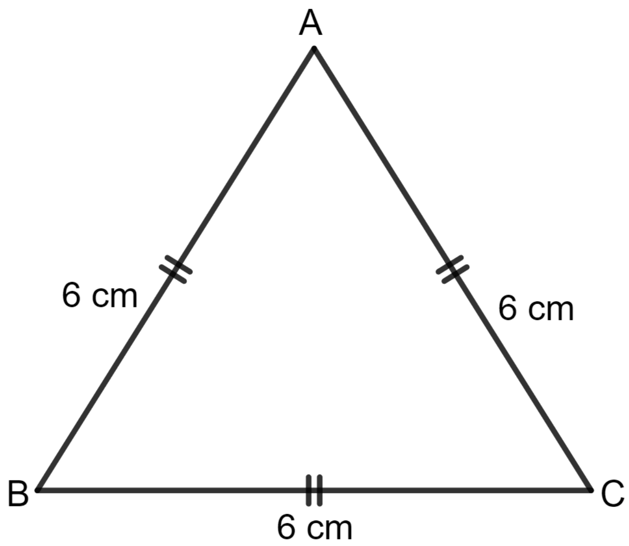
3. $18\sqrt{3}$ cm^2

4. $12\sqrt{3}$ cm^2

Answer

Given:

Side of the equilateral triangle = 6 cm



$\because$ As we know, the area of the equilateral triangle = $\dfrac{\sqrt{3}}{4} \times \text{side}^2$

= $\dfrac{\sqrt{3}}{4} \times 6^2$ cm^2

= $\dfrac{\sqrt{3}}{4} \times 36$ cm^2

= $9\sqrt{3}$ cm^2

Hence, option 2 is the correct option.

Question 2

Find the area of a triangle whose sides are:

10 cm, 24 cm and 26 cm

Answer

Let a = 10 cm, b = 24 cm and c = 26 cm.

$$\because s = \frac{a + b + c}{2}$$

$$= \frac{10 + 24 + 26}{2}$$

$$= \frac{60}{2}$$

$$= 30$$

∵ Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

= $\sqrt{30(30-10)(30-24)(30-26)}$ cm^2

= $\sqrt{30 \times 20 \times 6 \times 4}$ cm^2

= $\sqrt{14,400}$ cm^2

= 120 cm^2

Hence, area of triangle = 120 cm^2.

Question 3

Two sides of a triangle are 6 cm and 8 cm. If the height of the triangle corresponding to 6 cm side is 4 cm; find :

(i) area of the triangle

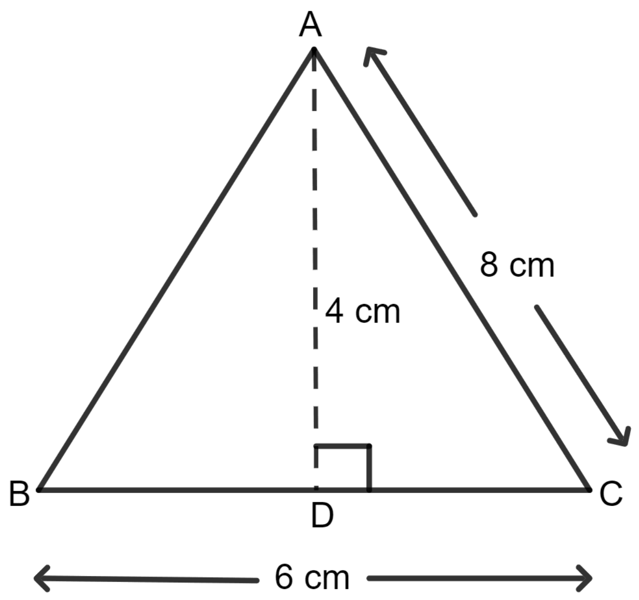
(ii) height of the triangle corresponding to 8 cm side.

Answer

(i) **Given:**

Side of the triangle = 6 cm

Height of the triangle = 4 cm



As we know, the area of the triangle = $\frac{1}{2}$ x side x height

= $\dfrac{1}{2}$ x 6 x 4 cm^2

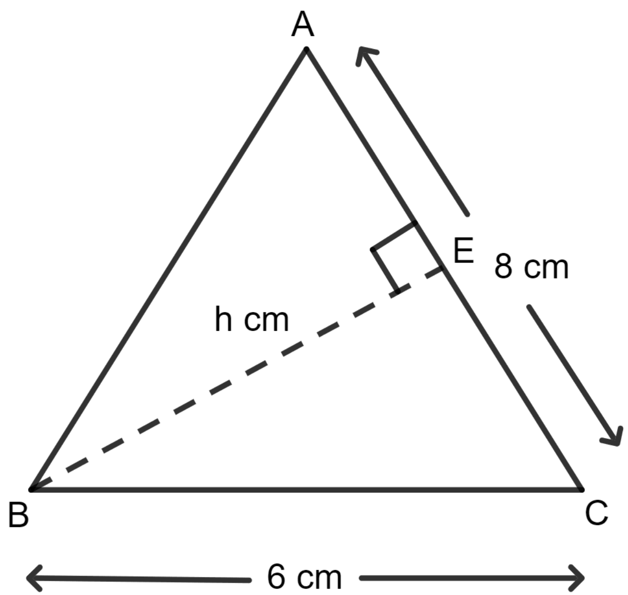
= $\dfrac{1}{2}$ x 24 cm^2

= 12 cm^2

Hence, area of triangle = 12 cm^2.

(ii) When, the side of the triangle = 8 cm

Let the height of the triangle be h cm



As we know area of triangle = $\dfrac{1}{2}$ x side x height

= $\dfrac{1}{2}$ x 8 x h ^{2}

[∵ Both areas remain same since they represent the area of the same triangle.]

⇒ $\dfrac{1}{2}$ x 8 x h = 12

⇒ 4 x h = 12

⇒ h = $\dfrac{12}{4}$

⇒ h = 3 cm

Hence, height of the triangle corresponding to 8 cm side is 3 cm.

Question 4

The sides of a triangle are 16 cm, 12 cm and 20 cm. Find:

(i) area of the triangle

(ii) height of the triangle corresponding to the largest side

(iii) height of the triangle corresponding to the smallest side.

Answer

(i) Let a = 16 cm, b = 12 cm and c = 20 cm.

$$\because s = \dfrac{a + b + c}{2}$$

$$= \dfrac{16 + 12 + 20}{2}$$

$$= \dfrac{48}{2}$$

$$= 24$$

∵ Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

= $\sqrt{24(24-16)(24-12)(24-20)}$ cm^2

= $\sqrt{24 \times 8 \times 12 \times 4}$ cm^2

= $\sqrt{9,216}$ cm^2

= 96 cm^2

Hence, area of triangle = 96 cm^2.

(ii) When, the largest side of the triangle = 20 cm

Let the height of the triangle = h cm

As we know, the area of the triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 20 x h

[∵ Both areas remain same since they represent the area of the same triangle.]

⇒ $\frac{1}{2}$ x 20 x h = 96

⇒ 10 x h = 96

⇒ h = $\frac{96}{10}$

⇒ h = 9.6

Hence, height of the triangle corresponding to 20 cm side is 9.6 cm.

(iii) When, the smallest side of the triangle = 12 cm

Let the height of the triangle = h cm

As we know, the area of the triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 12 x h

[∵ Both areas remain same since they represent the area of the same triangle.]

⇒ $\frac{1}{2}$ x 12 x h = 96

⇒ 6 x h = 96

⇒ h = $\frac{96}{6}$

⇒ h = 16 cm

Hence, height of the triangle corresponding to 12 cm side is 16 cm.

Question 5

The base and the height of a triangle are in the ratio 4 : 5. If the area of the triangle is 40 m^2, find its base and height.

Answer

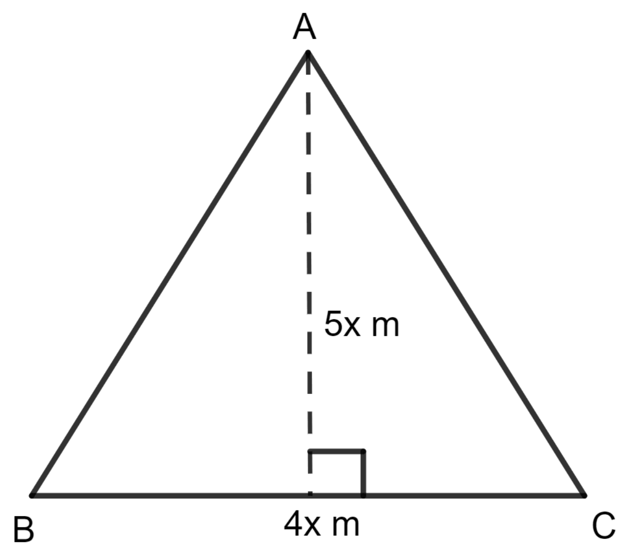
Given:

The base and the height of a triangle are in the ratio 4 : 5.

The area of the triangle = 40 m^2

Let the base of the triangle be 4x m.

And, the height of the triangle be 5x m.



As we know, the area of a triangle = $\frac{1}{2} \times$ side $\times$ height

$\Rightarrow \frac{1}{2} \times 4x \times 5x = 40 \text{ m}^2$

$\Rightarrow \frac{1}{2} \times 20x^2 = 40 \text{ m}^2$

$\Rightarrow 10x^2 = 40 \text{ m}^2$

$\Rightarrow x^2 = \frac{40}{10} \text{ m}^2$

$\Rightarrow x^2 = 4 \text{ m}^2$

$\Rightarrow x = \sqrt{4}$ m

⇒ x = 2 m

So, base of the triangle = 4x

= 4 x 2 m

= 8 m

And, height of the triangle = 5x

= 5 x 2 m

= 10 m

Hence, base and height of the triangle are 8 m and 10 m, respectively.

Question 6

The base and the height of a triangle are in the ratio 5 : 3. If the area of the triangle is 67.5 m^2; find its base and height.

Answer

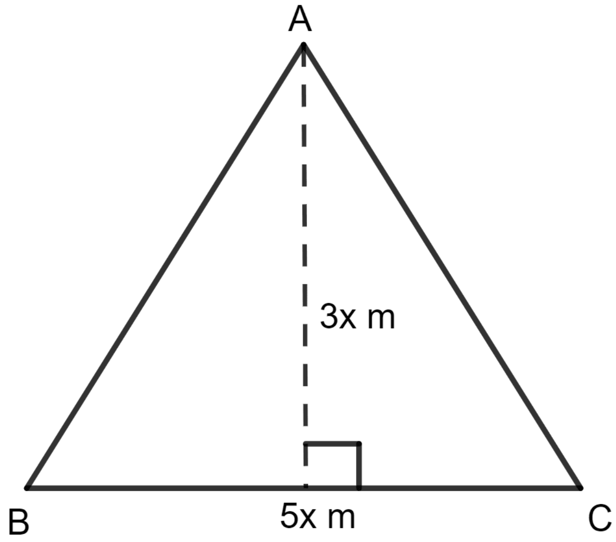
Given:

The base and the height of a triangle are in the ratio 5 : 3.

The area of the triangle = 67.5 m^2

Let the base of the triangle = 5x m

And, the height of the triangle = 3x m



As we know, the area of the triangle = $\frac{1}{2}$ x side x height

⇒ $\frac{1}{2}$ x 5x x 3x = 67.5 m^2

⇒ $\frac{1}{2}$ x $15x^2$ = 67.5 m^2

⇒ $15x^2 = 2 \times 67.5$ m^2

⇒ $15x^2$ = 135 m^2

⇒ $x^2 = \frac{135}{15}$ m^2

⇒ x^2 = 9 m^2

⇒ x = $\sqrt{9}$ m

⇒ x = 3 m

So, base of the triangle = 5x

= 5 x 3 m

= 15 m

And, height of the triangle = 3x

= 3 x 3 m

= 9 m

Hence, base and height of the triangle are 15 m and 9 m, respectively.

Question 7

The area of an equilateral triangle is $144\sqrt{3}\text{ cm}^2$; find its perimeter.

Answer

Let the side of the equilateral triangle be a .

It is given that the area of an equilateral triangle is $144\sqrt{3}\text{ cm}^2$.

∵ As we know, the area of the equilateral triangle = $\dfrac{\sqrt{3}}{4} \times \text{side}^2$

$\Rightarrow \dfrac{\sqrt{3}}{4} \times a^2 = 144\sqrt{3}\text{ cm}^2$

$\Rightarrow \dfrac{\cancel{\sqrt{3}}}{4} \times a^2 = 144\cancel{\sqrt{3}}\text{ cm}^2$

$\Rightarrow \dfrac{1}{4} \times a^2 = 144\text{ cm}^2$

$\Rightarrow a^2 = 4 \times 144\text{ cm}^2$

⇒ $a^2 = 576 \text{ cm}^2$

⇒ $a = \sqrt{576} \text{ cm}^2$

⇒ a = 24 cm

Now, perimeter of equilateral triangle = 3 x side

= 3 x 24 cm

= 72 cm

Hence, the perimeter of the triangle is 72 cm.

Question 8

The area of an equilateral triangle is numerically equal to its perimeter. Find its perimeter correct to 2 decimal places.

Answer

Let the side of the equilateral triangle be `a` .

∵ As we know, the area of the equilateral triangle = $\dfrac{\sqrt{3}}{4} \times \text{ side}^2$

And, perimeter of equilateral triangle = 3 x side

It is given that the area of an equilateral triangle is numerically equal to its perimeter.

⇒ $\dfrac{\sqrt{3}}{4}a^2 = 3a$

⇒ $\sqrt{3}a^2 = 4 \times 3a$

⇒ $\sqrt{3}a^2 = 12a$

⇒ $\sqrt{3}a = 12$

⇒ $a = \dfrac{12}{\sqrt{3}}$

⇒ $a = \dfrac{12 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$

⇒ $a = \dfrac{12\sqrt{3}}{3}$

⇒ $a = 4\sqrt{3}$

As we know perimeter of equilateral triangle = 3 x side

= 3 x $4\sqrt{3}$ unit

= 12 $\sqrt{3}$ unit

= 12 x 1.74 unit

= 20.78 unit

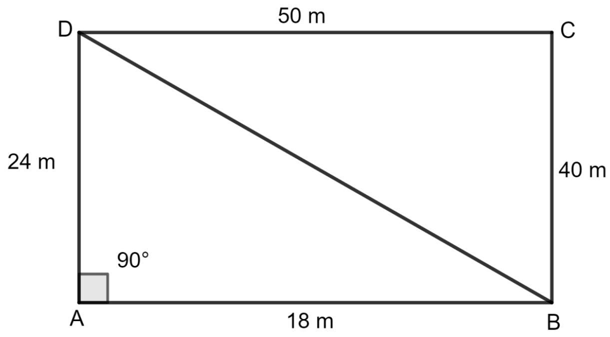
Hence, the perimeter of the triangle is 20.78 unit.

Question 9

A field is in the shape of a quadrilateral ABCD in which side AB = 18 m, side AD = 24 m, side BC = 40 m, DC = 50 m and angle A = 90°. Find the area of the field.

Answer

Join BD and the field is divided into two triangular fields.



Triangle ABD is a right angled triangle.

AB = 18 m

AD = 24 m

Let BD be h m.

By using the Pythagorean theorem,

Base^2 + Height^2 = Hypotenuse^2

$\Rightarrow (18)^2 + (24)^2 = h^2$

$\Rightarrow 324 + 576 = h^2$

$\Rightarrow h^2 = 900$

$\Rightarrow h = \sqrt{900}$

$\Rightarrow$ h = 30 m

For the right-angled triangle ABD,

As we know, the area of a triangle = $\dfrac{1}{2}$ x base x height

= $\dfrac{1}{2}$ x 18 x 24 m^2

= $\dfrac{1}{2}$ x 432 m^2

= 216 m^2

For triangle BCD,

Let a = 40 m, b = 50 m and c = 30 m.

$$s = \frac{a + b + c}{2}$$

$$= \frac{40 + 50 + 30}{2}$$

$$= \frac{120}{2}$$

$$= 60$$

∵ Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

= $\sqrt{60(60-40)(60-50)(60-30)}$ m^2

= $\sqrt{60 \times 20 \times 10 \times 30}$ m^2

= $\sqrt{3,60,000}$ m^2

= 600 m^2

Area of rectangular field = Area of triangle ABD + Area of triangle BCD

= 216 + 600 m^2

= 816 m^2

Hence, the area of the field is 816 m^2.

Question 10

The lengths of the sides of a triangle are in the ratio 4 : 5 : 3 and its perimeter is 96 cm. Find its area.

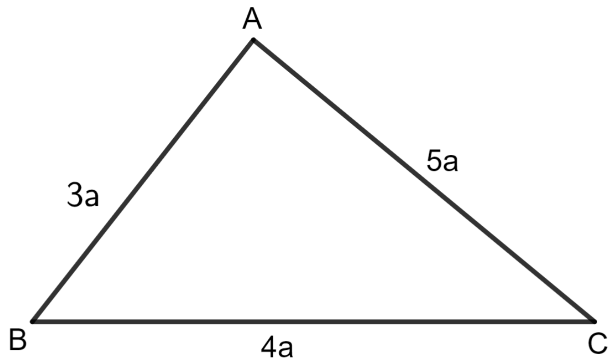
Answer

Given:

Perimeter of the triangle = 96 cm

The lengths of the sides of a triangle are in the ratio 4 : 5 : 3.

Let the length of the sides of the triangle be 4a, 5a and 3a.



∵ Perimeter of triangle = sum of sides of the triangle.

⇒ 4a + 5a + 3a = 96 cm

⇒ 12a = 96 cm

⇒ a = $\dfrac{96}{12}$ cm

⇒ a = 8 cm

So, the lengths of the sides are 4a, 5a and 3a.

= 4 x 8, 5 x 8 and 3 x 8

= 32, 40 and 24

Let a = 32 m, b = 40 m and c = 24 m.

$\because s = \dfrac{a + b + c}{2}$

$= \dfrac{32 + 40 + 24}{2}$

$= \dfrac{96}{2}$

$= 48$

∵ Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

= $\sqrt{48(48-32)(48-40)(48-24)}$ cm^2

= $\sqrt{48 \times 16 \times 8 \times 24}$ cm^2

= $\sqrt{1,47,456}$ cm^2

= 384 cm^2

Hence, the area of the triangle is 384 cm^2.

Question 11

One of the equal sides of an isosceles triangle is 13 cm and its perimeter is 50 cm. Find the area of the triangle.

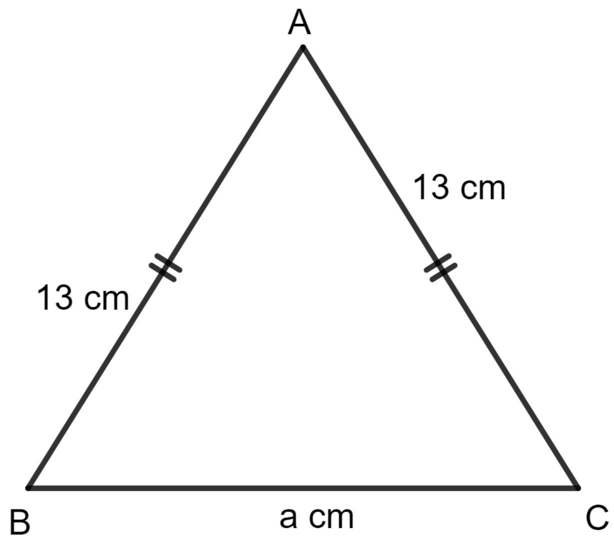
Answer

Given:

One of the equal sides of an isosceles triangle = 13 cm

Perimeter of the triangle = 50 cm

Let a be the length of the third side of the triangle.



∵ Perimeter of triangle = sum of sides

⇒ 13 + 13 + a = 50 cm

⇒ 26 + a = 50 cm

⇒ a = 50 - 26 cm

⇒ a = 24 cm

Let a = 13 m, b = 13 m and c = 24 m.

$\because s = \dfrac{a + b + c}{2}$

$= \dfrac{13 + 13 + 24}{2}$

$= \dfrac{50}{2}$

$= 25$

∵ Area of triangle = $\sqrt{s(s - a)(s - b)(s - c)}$

= $\sqrt{25(25 - 13)(25 - 13)(25 - 24)}$ cm^2

= $\sqrt{25 \times 12 \times 12 \times 1}$ cm^2

= $\sqrt{3,600}$ cm^2

= 60 cm^2

Hence, the area of the triangle is 60 cm^2.

Question 12

The altitude and the base of a triangular field are in the ratio 6 : 5. If its cost is ₹ 49,57,200 at the rate of ₹ 36,720 per hectare and 1 hectare = 10,000 sq.m, find (in metre) the dimensions of the field.

Answer

Given:

Ratio of altitude and base of the field = 6 : 5

Total cost = ₹ 49,57,200

Rate = ₹ 36,720 per hectare

As we know that area of the triangular field x rate = total cost

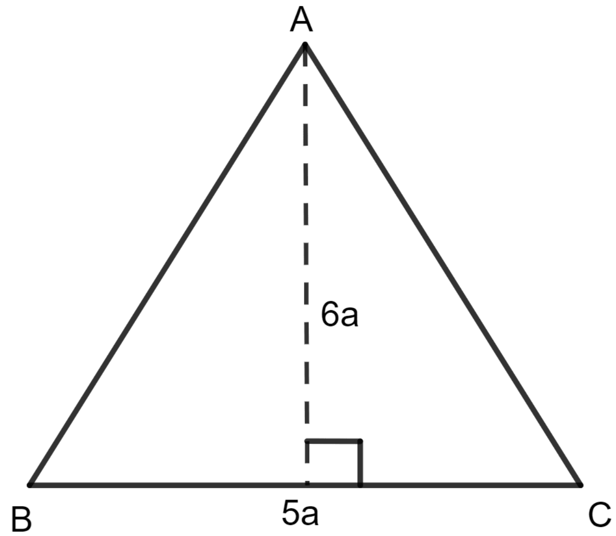
⇒ Area of the triangular field x $\dfrac{₹36,720}{1,00,00}$ = ₹ 49,57,200

⇒ Area of the triangular field = ₹ $\dfrac{49,57,200}{36,720} \times 10,000$

⇒ Area of the triangular field = ₹ $135 \times 10,000$

⇒ Area of the triangular field = 13,50,000 m^2

Let the altitude be 6a and the base be 5a.



As we know, the area of a triangle = $\frac{1}{2}$ x base x height

⇒ 13,50,000 = $\frac{1}{2}$ x 6a x 5a

⇒ 13,50,000 = $\frac{1}{2}$ x $30a^2$

⇒ 13,50,000 = $15a^2$

⇒ $a^2 = \frac{13,50,000}{15}$

⇒ a^2 = 90000

⇒ a = $\sqrt{90000}$

⇒ a = 300

So, the altitude = 6a

= 6 x 300

= 1800 m

And, the base = 5a

= 5 x 300

= 1500 m

Hence, the dimensions of the field are 1800 m and 1500 m.

Question 13

Find the area of the right-angled triangle with hypotenuse 40 cm and one of the other two sides 24 cm.

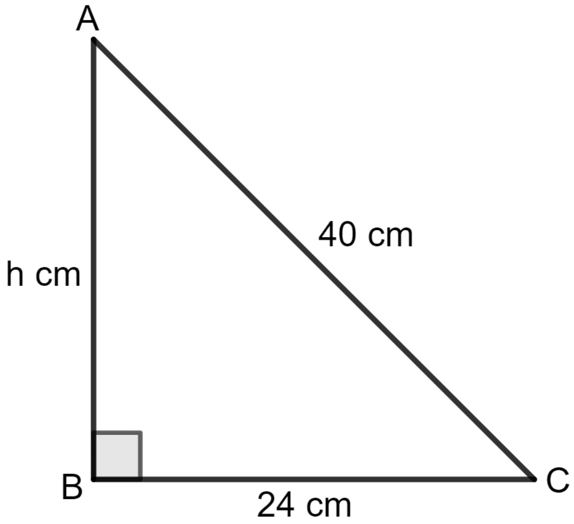
Answer

Given:

Base of the triangle = 24 cm

Hypotenuse of the triangle = 40 cm

Let h be the height of the triangle.



In a right-angled triangle, using the Pythagorean theorem,

$\text{Base}^2 + \text{Height}^2 = \text{Hypotenuse}^2$

$\Rightarrow (24)^2 + h^2 = (40)^2$

$\Rightarrow 576 + h^2 = 1600$

$\Rightarrow h^2 = 1600 - 576$

$\Rightarrow h^2 = 1024$

$\Rightarrow h = \sqrt{1024}$

$\Rightarrow h = 32$ cm

As we know, the area of a triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 24 x 32 cm^2

= $\dfrac{1}{2}$ x 768 cm^2

= 384 cm^2

Hence, the area of the triangle is 384 cm^2.

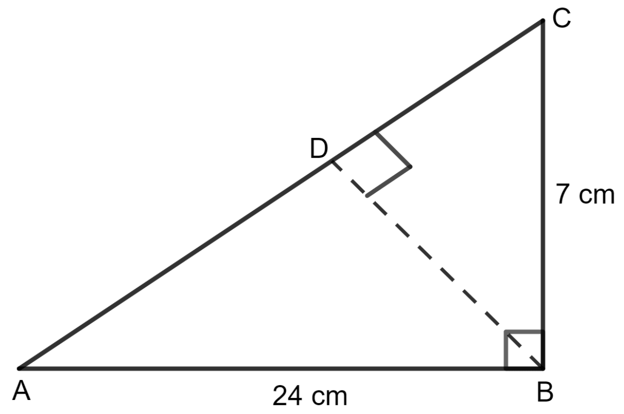
Question 14

Use the information given in the given figure to find:

(i) the length of AC.

(ii) the area of △ ABC.

(iii) the length of BD, correct to one decimal place.



Answer

(i) **Given:**

In triangle ABC,

Base of the triangle = 24 cm

Height of the triangle = 7 cm

Let h be the hypotenuse of the triangle.

In a right-angled triangle, using the Pythagorean theorem,

$\text{Base}^2 + \text{Height}^2 = \text{Hypotenuse}^2$

$\Rightarrow (24)^2 + (7)^2 = h^2$

$\Rightarrow 576 + 49 = h^2$

$\Rightarrow h^2 = 625$

$\Rightarrow h = \sqrt{625}$

$\Rightarrow$ h = 25 cm

Hence, the length of AC = 25 cm.

(ii) As we know, the area of a triangle = $\frac{1}{2}$ x side x height

= $\frac{1}{2}$ x 24 x 7 cm^2

= $\frac{1}{2}$ x 168 cm^2

= 84 cm^2

Hence, the area of the triangle = 84 cm^2.

(iii) The base of the triangle = 25 cm

Let the height of the triangle be h .

As we know, the area of a triangle = $\frac{1}{2}$ x side x height

[∵ Both areas remain same since they represent the area of the same triangle.]

⇒ $\frac{1}{2}$ x 25 x h = 84

⇒ 25 x h = 2 x 84

⇒ 25 x h = 168

⇒ h = $\frac{168}{25}$

⇒ h = 6.7 cm

Hence, the length of BD = 6.7 cm.

Exercise 22(B)

Question 1(i)

The length and the breadth of a rectangle are in the ratio 9 : 5. If its area is 180 m^2; the length (longest side) of the rectangle is :

1. 36 m
2. 20 m
3. 10 m
4. 18 m

Answer

Given:

The length and the breadth of a rectangle are in the ratio 9 : 5.

The area of the rectangle is 180 m^2.

Let the length of the rectangle be 9a and the breadth of the rectangle be 5a.

As we know, the area of the rectangle = length x breadth

⇒ 180 m^2 = 9a x 5a

⇒ 180 m^2 = $45a^2$

⇒ $a^2 = \frac{180}{45}$ m^2

⇒ a^2 = 4 m^2

⇒ a = $\sqrt{4}$ m

⇒ a = 2 m

So, the length of the rectangle is 9a.

= 9 x 2 m

= 18 m

Hence, option 4 is the correct option.

Question 1(ii)

The adjacent sides of a rectangle are 16 m and 9 m. The length of the square whose area is equal to the area of the rectangle is :

1. 16 m
2. 18 m
3. 12 m
4. 72 m

Answer

Given:

The adjacent sides of the rectangle are 16 m and 9 m.

As we know, the area of the rectangle = length x breadth

= 16 x 9 m^2

= 144 m^2

SInce the area of the rectangle is equal to the area of the square, And, the area of the square = side^2

⇒ side^2 = 144 m^2

⇒ side = $\sqrt{144}$ m

⇒ side = 12 m

Hence, option 3 is the correct option.

Question 1(iii)

The area of a square, with perimeter 16 cm is :

1. 16 cm^2
2. 8 cm^2
3. 12 cm^2
4. 128 cm^2

Answer

Given:

The perimeter of the square is 16 cm.

As we know, the perimeter of a square = 4 x side.

⇒ 4 x side = 16 cm

⇒ side = $\dfrac{16}{4}$ cm

⇒ side = 4 cm

And, the area of the square = side^2

= $(4)^2$ cm^2

= 16 cm^2

Hence, option 1 is the correct option.

Question 1(iv)

The perimeter of square with area 169 m^2 is :

1. 52 m
2. 26 m
3. 39 m
4. 19.5 m^2

Answer

Given:

The area of the square is 169 m^2.

Let the side of the square be x m.

As we know, the area of a square = side^2

⇒ x^2 = 169 m^2

⇒ x = $\sqrt{169}$ m

⇒ x = 13 m

And, the perimeter of the square = 4 x side

= 4 x 13 m

= 52 m

Hence, option 1 is the correct option.

Question 1(v)

The area of square with diagonal 10 cm is:

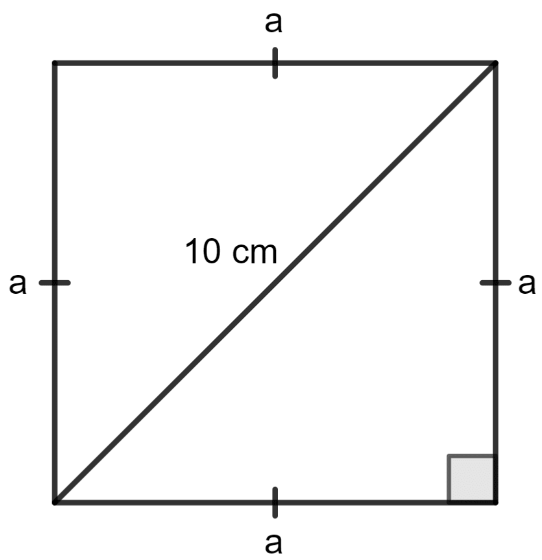
1. 100 cm^2
2. 50 cm^2
3. 25 cm^2
4. 75 cm^2

Answer

Given:

The diagonal of the square is 10 cm.

Let each side of the square be a cm.



Then, by using the Pythagoras theorem,

⇒ Diagonal^2 = Side^2 + Side^2

⇒ $(10)^2 = a^2 + a^2$

⇒ $100 = 2a^2$

⇒ $a^2 = \frac{100}{2}$

⇒ $a^2 = 50$

⇒ $a = \sqrt{50}$

⇒ $a = 5\sqrt{2}$

As we know, the area of a square = side^2

= $(5\sqrt{2})^2$ cm^2

= 25 x 2 cm^2

= 50 cm^2

Hence, option 2 is the correct option.

Question 2

Find the length and perimeter of a rectangle, whose area = 120 cm^2 and breadth = 8 cm.

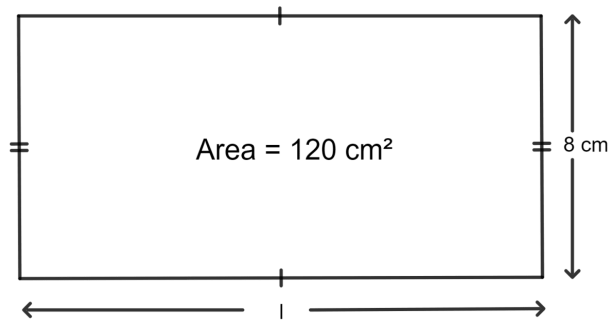
Answer

Given:

The breadth of the rectangle is 8 cm.

The area of the rectangle is 120 cm^2.

Let the length of the rectangle be l.



As we know, the area of the rectangle = length x breadth

⇒ l x 8 = 120

⇒ l = $\dfrac{120}{8}$

⇒ l = 15 cm

The perimeter of the rectangle = 2(length + breadth)

= 2 (15 + 8)

= 2 x 23

= 46 cm

Hence, the length of the rectangle is 15 cm, and the perimeter is 46cm.

Question 3

The perimeter of a rectangle is 46 m and its length is 15 m. Find its :

(i) breadth

(ii) area

(iii) diagonal

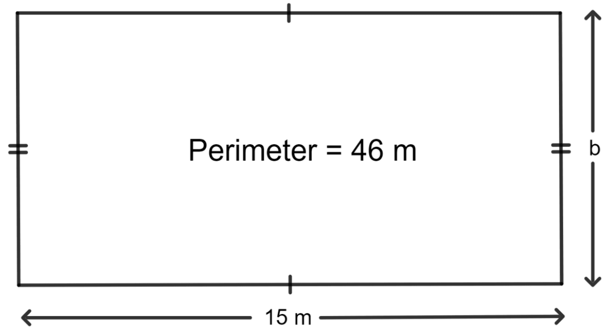
Answer

(i) **Given:**

The perimeter of the rectangle is 46 m.

The length of the rectangle is 15 m.

Let the breadth of the rectangle be b m.



As we know, the perimeter of the rectangle = 2(length + breadth)

⇒ 2(15 + b) = 46

⇒ 15 + b = $\dfrac{46}{2}$

⇒ 15 + b = 23

⇒ b = 23 - 15

⇒ b = 8 m

Hence, the breadth of the rectangle is 8 m.

(ii) The area of the rectangle = length x breadth

= 15 x 8 m^2

= 120 m^2

Hence, the area of the rectangle is 120 m^2.

(iii) By using Pythagoras theorem,

⇒ $\text{Diagonal}^2 = \text{Length}^2 + \text{Breadth}^2$

⇒ Diagonal^2 = 15^2 + 8^2

⇒ Diagonal^2 = 225 + 64

⇒ Diagonal^2 = 289

⇒ Diagonal = $\sqrt{289}$

⇒ Diagonal = 17 m

Hence, the diagonal of the rectangle is 17 m.

Question 4

The diagonal of a rectangle is 34 cm. If its breadth is 16 cm, find its :

(i) length

(ii) area

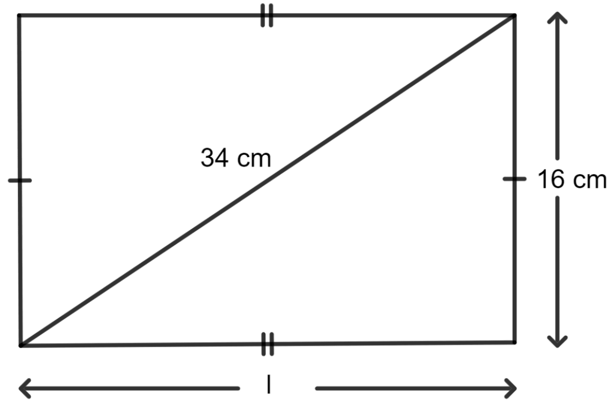
Answer

Given:

The diagonal of the rectangle is 34 cm.

The breadth of the rectangle is 16 cm.

Let the length of the rectangle be `l` cm.



By using Pythagoras theorem,

⇒ $\text{Diagonal}^2 = \text{Length}^2 + \text{Breadth}^2$

⇒ $(34)^2 = l^2 + (16)^2$

⇒ $1,156 = l^2 + 256$

⇒ $l^2 = 1,156 - 256$

⇒ $l^2 = 900$

⇒ $l = \sqrt{900}$

⇒ l = 30 cm

Hence, the length of the rectangle is 30 cm.

(ii) As we know, the area of the rectangle = length x breadth

= 30 x 16 cm^2

= 480 cm^2

Hence, the area of the rectangle is 480 cm^2.

Question 5

The area of a small rectangular plot is 84 m^2. If the difference between its length and the breadth is 5 m, find its perimeter.

Answer

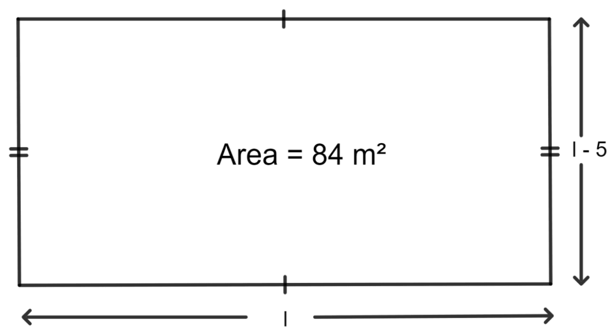
Given:

The area of a small rectangular plot is 84 m^2.

The difference between its length and the breadth is 5 m.

Let the length of the rectangle be l m.

So, breadth of the rectangle = l - 5



As we know, the area of the rectangle = length x breadth

⇒ l x (l - 5) = 84

⇒ l^2 - 5l = 84

$\Rightarrow l^2 - 5l - 84 = 0$

$\Rightarrow l^2 - (12 - 7)l - 84 = 0$

$\Rightarrow l^2 - 12l + 7l - 84 = 0$

$\Rightarrow l(l - 12) + 7(l - 12) = 0$

$\Rightarrow (l - 12)(l + 7) = 0$

$\Rightarrow l = 12 \text{ or } - 7$

Since length cannot be negative, the length will be 12 m.

So, breadth of the rectangle = $(l - 5)$

= (12 - 5) m

= 7 m

And, perimeter of the rectangle = 2(length + breadth)

= 2(12 + 7) m

= 2 x 19 m

= 38 m

Hence, the perimeter of the rectangle is 38 m.

Question 6

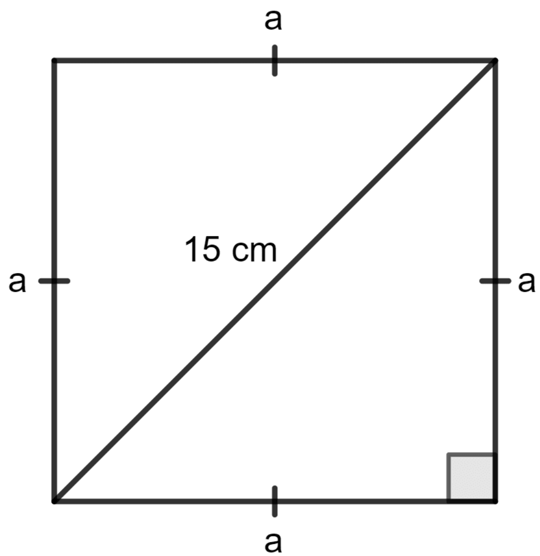
The diagonal of a square is 15 m; find the length of its one side and perimeter.

Answer

Given:

Diagonal of the square is 15 m.

Let each side of the square be a m.



Then, by using Pythagoras theorem,

⇒ Diagonal^2 = Side^2 + Side^2

⇒ $(15)^2 = a^2 + a^2$

⇒ $225 = 2a^2$

⇒ $a^2 = \dfrac{225}{2}$

⇒ $a^2 = 112.5$

⇒ $a = \sqrt{112.5}$

⇒ a = 10.6

As we know that perimeter of the square = 4 x side

= 4 x 10.6 m

= 42.4 m

Hence, the side of square is 10.6 m and its perimeter is 42.4 m.

Question 7

The length of a rectangle is 16 cm and its perimeter is equal to the perimeter of a square with side 12.5 cm. Find the area of the rectangle.

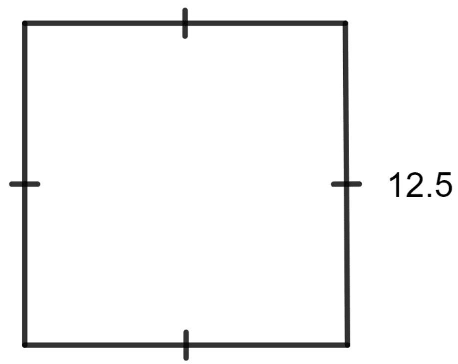
Answer

Given:

The length of the rectangle is 16 cm.

The perimeter of the rectangle = The perimeter of a square.

The side of the square is 12.5 cm.

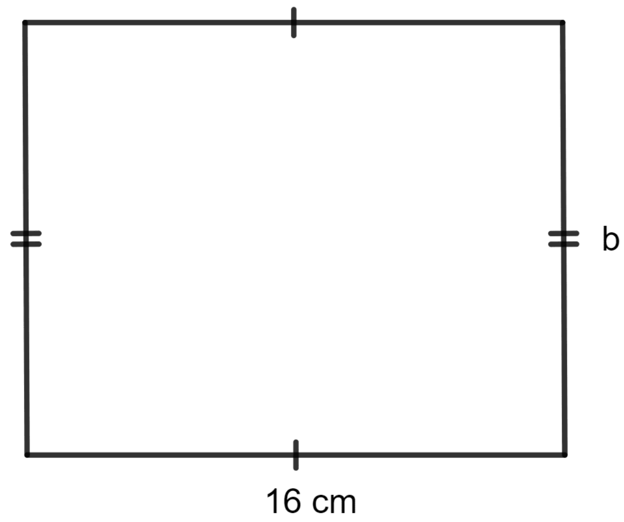


As we know, the perimeter of the square = 4 x side

= 4 x 12.5 cm

= 50 cm

Let the breadth of the rectangle be b cm.



And, the perimeter of the rectangle = 2 (length + breadth)

⇒ 2 (16 + b) = 50

⇒ 16 + b = $\frac{50}{2}$

⇒ 16 + b = 25

⇒ b = 25 - 16

⇒ b = 9 cm

Now, area of the rectangle = length x breadth

= 16 x 9 cm^2

= 144 cm^2

Hence, the area of the rectangle is 144 cm^2.

Question 8

The perimeter of a square is numerically equal to its area. Find its area.

Answer

Given:

The perimeter of a square = The area of a square.

Let the side of the square be a unit.

As we know, the perimeter of the square = 4 x side

= 4a

And, the area of the square = $side^2$

= a^2

Thus, 4a = a^2

⇒ a = 4 unit

Now, the area of the square = a^2

= 4^2

= 16 sq. units

Hence, the area of the square is 16 sq. units.

Question 9

Each side of a rectangle is doubled. Find the ratio between :

(i) perimeters of the original rectangle and the resulting rectangle

(ii) areas of the original rectangle and the resulting rectangle

Answer

(i) **Given:**

Each side of a rectangle is doubled.

Let the length and breadth of the rectangle be l and b.

Thus, the new length and breadth of the rectangle are 2l and 2b.

As we know, the perimeter of a rectangle = 2 (length + breadth)

The perimeter of the original rectangle = 2 (l + b)

The perimeter of the new rectangle = 2 (2l + 2b)

= 4 (l + b)

Thus, the ratio of the perimeters of the original rectangle and the resulting rectangle = $\dfrac{2(l+b)}{4(l+b)}$

= $\dfrac{2\cancel{(l+b)}}{4\cancel{(l+b)}}$

= $\dfrac{2}{4}$

= $\dfrac{1}{2}$

Hence, the ratio of the perimeters of the original rectangle and the resulting rectangle is $\dfrac{1}{2}$.

(ii) As we know, the area of a rectangle = length x breadth

The area of the original rectangle = l x b

= lb

The area of the new rectangle = 2l x 2b

= 4 lb

Thus, the ratio of the areas of the original rectangle and the resulting rectangle

= $\dfrac{lb}{4lb}$

= $\dfrac{\cancel{lb}}{4\cancel{lb}}$

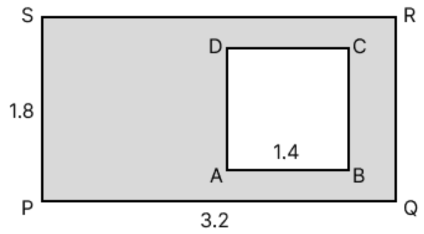
= $\dfrac{1}{4}$

Hence, the ratio of areas of the original rectangle and the resulting rectangle is $\dfrac{1}{4}$.

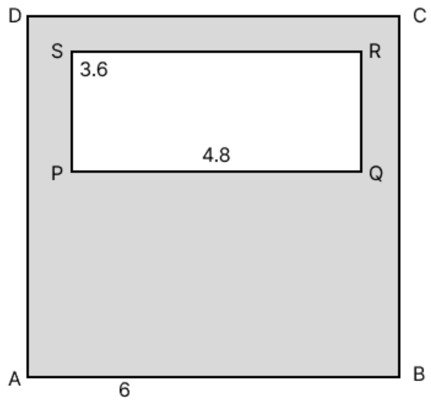
Question 10

In each of the following cases ABCD is a square and PQRS is a rectangle. Find, in each case, the area of the shaded portion. (All measurements are in metre).

(i)



(ii)



Answer

(i) The area of the shaded portion = Area of the rectangle - Area of the square.

It is given that the length and breadth of the rectangle area 3.2 m and 1.8 m.

As we know, the area of the rectangle = length x breadth

= 3.2 x 1.8 m^2

= 5.76 m^2

The side of the square is 1.4 m.

And, the area of the square = $side^2$

= 1.4^2 m^2

= 1.96 m^2

Thus, the area of the shaded portion = 5.76 - 1.96 m^2

= 3.8 m^2

Hence, the area of the shaded portion is 3.8 m^2.

(ii) The area of the shaded portion = Area of the square - Area of the rectangle.

It is given that the side of the square is 6 m.

And, the area of the square = $side^2$

= 6^2 m^2

= 36 m^2

The length and breadth of the rectangle are 4.8 m and 3.6 m, respectively.

As we know, the area of the rectangle = length x breadth

= 4.8 x 3.6 m^2

= 17.28 m^2

Area of the shaded portion = 36 - 17.28 m^2

= 18.72 m^2

Hence, the area of the shaded portion is 18.72 m^2.

Question 11

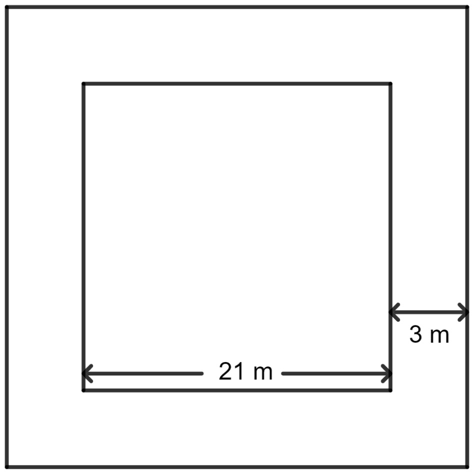
A path of uniform width, 3 m, runs around the outside of a square field of side 21 m. Find the area of the path.

Answer

Given:

The side of the square field is 21 m.

The width of the path is 3 m.



Area of the path = Area of bigger square field - Area of smaller square field

The side of the bigger square = 21 m + 3 m + 3m

= 27 m

As we know, the area of square = side^2

⇒ Area of the bigger square = 27^2 m^2

= 729 m^2

⇒ Area of the smaller square = 21^2 m^2

= 441 m^2

So, the area of the path = 729 - 441 m^2

= 288 m^2

Hence, the area of the path is 288 m^2.

Question 12

A path of uniform width, 2.5 m, runs around the inside of a rectangular field 30 m by 27 m. Find the area of the path.

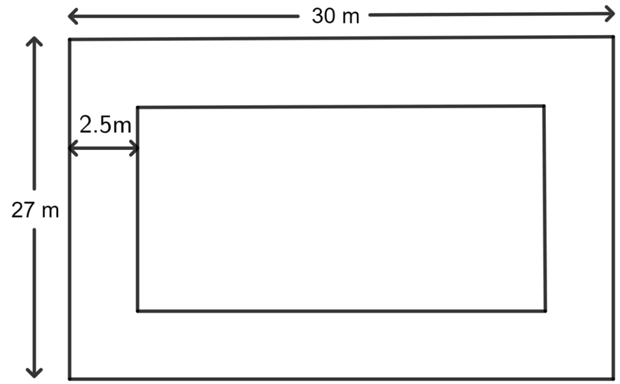
Answer

Given:

The length of the rectangular field is 30 m.

The breadth of the rectangular field is 27 m.

The width of the path is 2.5 m.



Area of the path = Area of bigger rectangular field - Area of smaller rectangular field

The length of the smaller rectangular field = 30 m - 2.5 m - 2.5 m

= 25 m

The breadth of the smaller rectangular field = 27 m - 2.5 m - 2.5 m

= 22 m

As we know, the area of a rectangle = length x breadth

⇒ Area of the bigger rectangular field = 30 x 27 m^2

= 810 m^2

⇒ Area of the smaller square = 25 x 22 m^2

= 550 m^2

So, the area of the path = 810 - 550 m^2

= 260 m^2

Hence, the area of the path is 260 m^2.

Question 13

The length of a hall is 18 m and its width is 13.5 m. Find the least number of square tiles, each of side 25 cm, required to cover the floor of the hall,

(i) without leaving any margin.

(ii) leaving a margin of width 1.5 m all around.

In each case, find the cost of the tiles required at the rate of ₹ 6 per tile.

Answer

(i) **Given:**

Length of the floor = 18 m

Breadth of the floor = 13.5 m

As we know, the area of the floor = length x breadth

= 18 x 13.5 m^2

= 243 m^2

Side of the square tile = 25 cm

= $\dfrac{25}{100}$ m

= $\dfrac{1}{4}$ m

As we know, the area of a tile = side^2

= $\left(\dfrac{1}{4}\right)^2$ m^2

= $\dfrac{1}{16}$ m^2

Area of square tiles x Number of tiles = Area of the floor

⇒ Number of tiles = $\dfrac{\text{Area of the floor}}{\text{Area of square tiles}}$

⇒ Number of tiles = $\dfrac{243}{\dfrac{1}{16}}$

⇒ Number of tiles = 243 x 16

⇒ Number of tiles = 3,888

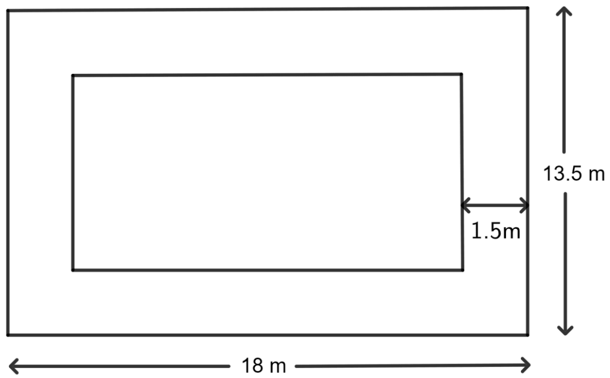
Rate per tile = ₹ 6 per tile

Total cost = ₹ 3,888 x 6

= ₹ 23,328

Hence, the number of tiles is 3,888 and the cost of the tiles is ₹ 23,328.

(ii) Width of the margin on each side = 1.5 m



Length of the inner floor = 18 m - 1.5 m - 1.5 m

= 18 - 3 m

= 15 m

Breadth of the inner floor = 13.5 m - 1.5 m - 1.5 m

= 13.5 - 3 m

= 10.5 m

As we know, the area of the floor = length x breadth

= 15 x 10.5 m^2

= 157.5 m^2

Side of the square tile = 25 cm

$= \frac{25}{100}$ m

$= \frac{1}{4}$ m

As we know, the area of a tile = side^2

= $\left(\dfrac{1}{4}\right)^2$ m^2

= $\dfrac{1}{16}$ m^2

Area of square tiles x Number of tiles = Area of the floor

⇒ Number of tiles = $\dfrac{\text{Area of the floor}}{\text{Area of square tiles}}$

⇒ Number of tiles = $\dfrac{157.5}{\dfrac{1}{16}}$

⇒ Number of tiles = 157.5 x 16

⇒ Number of tiles = 2,520

Rate of tiles = ₹ 6 per tile

Total cost = ₹ 2,520 x 6

= ₹ 15,120

Hence, the number of tiles is 2,520 and the cost of the tiles is ₹ 15,120.

Question 14

A rectangular field is 30 m in length and 22 m in width. Two mutually perpendicular roads, each 2.5 m wide, are drawn inside the field so that one road is parallel to the length of the field and the other road is parallel to its width. Calculate the area of the crossroads.

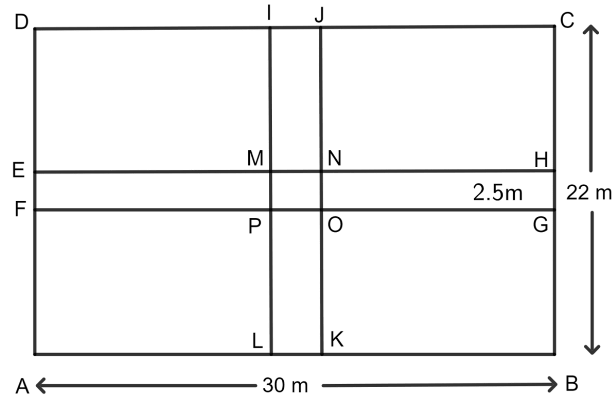
Answer

Given:

Length of the rectangular field = 30 m

Width of the rectangular field = 22 m

Width of each road = 2.5 m



Area of crossroad IJKL (parallel to the width of field) = 2.5 x 22 m^2

= 55 m^2

Area of crossroad EFGH (parallel to the length of field) = 2.5 x 30 m^2

= 75 m^2

Side of the square MNOP = 2.5

As we know, the area of the square = $side^2$

Area of the square field MNOP (area common to both roads) = 2.5^2 m^2

= 6.25 m^2

Now, area of crossroad = Area of cross road EFGH + Area of cross road IJKL - Area of square field MNOP

= 75 m^2 + 55 m^2 - 6.25 m^2

= 130 m^2 - 6.25 m^2

= 123.75 m^2

Hence, the area of the crossroads is 123.75 m^2.

Question 15

The length and the breadth of a rectangular field are in the ratio 5 : 4 and its area is 3380 m^2. Find the cost of fencing it at the rate of ₹ 75 per m.

Answer

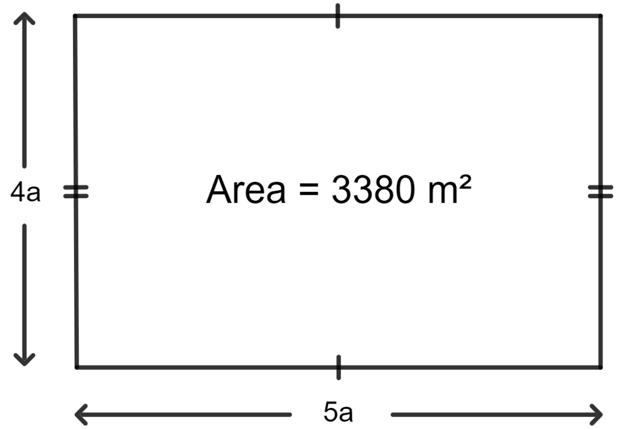
Given:

The length and the breadth of a rectangular field are in the ratio 5 : 4.

The area of the rectangular field = 3380 m^2

Rate of fencing = ₹ 75 per meter

Let the length of the rectangular field be 5a and the breadth be 4a.



As we know, the area of the rectangle = length x breadth

⇒ 5a x 4a = 3380 m^2

⇒ $20a^2$ = 3380 m^2

⇒ $a^2 = \frac{3380}{20}$ m^2

⇒ a^2 = 169 m^2

⇒ a = $\sqrt{169}$ m

⇒ a = 13 m

So, the length of the rectangular field = 5a

= 5 x 13 m

= 65 m

And, the breadth of the rectangular field = 4a.

= 4 x 13 m

= 52 m

Perimeter of the rectangle = 2(l + b)

= 2(65 + 52)

= 2 x 117

= 234 m

Total cost of fencing = Rate of fencing x Perimeter of fencing

= ₹ 75 x 234

= ₹ 17,550

Hence, the cost of fencing is ₹ 17,550.

Question 16

The length and the breadth of a conference hall are in the ratio 7 : 4 and its perimeter is 110 m. Find :

(i) area of the floor of the hall.

(ii) number of tiles, each a rectangle of size 25 cm x 20 cm, required for flooring of the hall.

(iii) the cost of the tiles at the rate of ₹ 1,400 per hundred tiles.

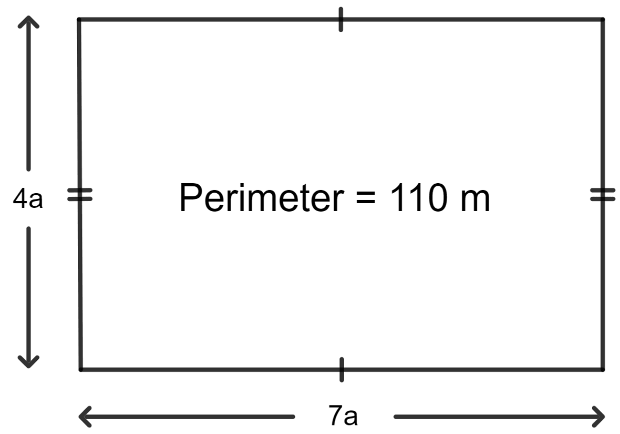
Answer

(i) **Given:**

The length and the breadth of a conference hall are in the ratio 7 : 4.

The perimeter of the hall = 110 m

Let the length of the conference hall be 7a and the breadth be 4a.



As we know, the perimeter of the hall = 2(length + breadth)

⇒ 2(7a + 4a) = 110

⇒ 2 x 11a = 110

⇒ 22a = 110

⇒ a = $\dfrac{110}{22}$

⇒ a = 5

So, the length of the hall = 7a

= 7 x 5 m

= 35 m

And, the breadth of the hall = 4a

= 4 x 5 m

= 20 m

Area of the hall = length x breadth

= 35 x 20 m^2

= 700 m^2

Hence, the area of the conference hall is 700 m^2.

(ii) **Given:**

Length of the tile = 25 cm

= $\dfrac{25}{100}$ m

= $\dfrac{1}{4}$ m

Breadth of the tile = 20 cm

= $\dfrac{20}{100}$ m

= $\dfrac{1}{5}$ m

Area of the tile = length x breadth

= $\dfrac{1}{4} \times \dfrac{1}{5}$ m^2

= $\dfrac{1}{20}$ m^2

Area of hall = Area of tiles x Number of tiles

⇒ 700 = $\dfrac{1}{20}$ x Number of tiles

⇒ Number of tiles = 700 x 20

⇒ Number of tiles = 14,000

Hence, the number of tiles = 14,000.

(iii) Cost of tiles = ₹ 1,400 per 100 tiles

Total cost = $\dfrac{14,000 \times 1400}{100}$

= ₹ 1,96,000

Hence, the cost of tiles = ₹ 1,96,000.

Exercise 22(C)

Question 1(i)

The parallel sides of a trapezium are 12 cm and 10 cm. If the distance between the parallel sides is 10 cm; the area of the trapezium is :

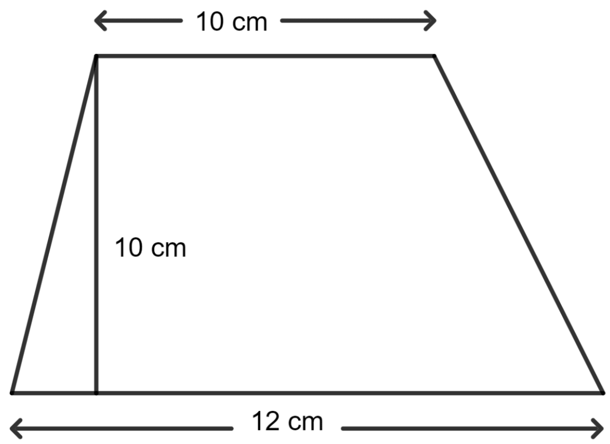
1. 220 cm^2
2. 110 cm^2
3. 55 cm^2
4. 165 cm^2

Answer

Given:

The lengths of the parallel sides of the trapezium are 12 cm and 10 cm.

The distance between the parallel sides is 10 cm.



As we know, the area of a trapezium = $\frac{1}{2}$ x (sum of parallel sides) x height

= $\frac{1}{2}$ x (12 + 10) x 10 cm^2

= $\frac{1}{2}$ x 22 x 10 cm^2

= $\frac{1}{2}$ x 220 cm^2

= 110 cm^2

Hence, option 2 is the correct option.

Question 1(ii)

Two parallel sides of a trapezium are in the ratio 2 : 3 and the distance between them is 12 cm. If the area of the trapezium is 240 cm^2; the length of its larger parallel side is :

1. 16 cm
2. 20 cm
3. 24 cm
4. 32 cm

Answer

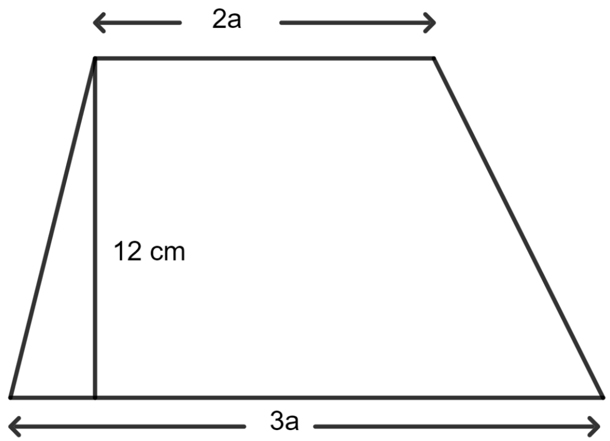
Given:

Two parallel sides of a trapezium are in the ratio 2 : 3.

The distance between the parallel sides = 12 cm.

The area of the trapezium = 240 cm^2.

Let the parallel sides of a trapezium be 2a and 3a.



As we know, the area of a trapezium = $\frac{1}{2}$ x (sum of parallel sides) x height

⇒ $\frac{1}{2}$ x (2a + 3a) x 12 cm^2 = 240 cm^2

⇒ $\frac{1}{2}$ x 5a x 12 = 240

⇒ $\frac{1}{2}$ x 60a = 240

⇒ 30a = 240

⇒ a = $\frac{240}{30}$

⇒ a = 8

So, the length of its larger parallel side = 3a

= 3 x 8 cm

= 24 cm

Hence, option 3 is the correct option.

Question 1(iii)

The adjacent sides of a parallelogram are 12 cm and 9 cm. The distance between longer sides is 6 cm; the distance between shorter sides is :

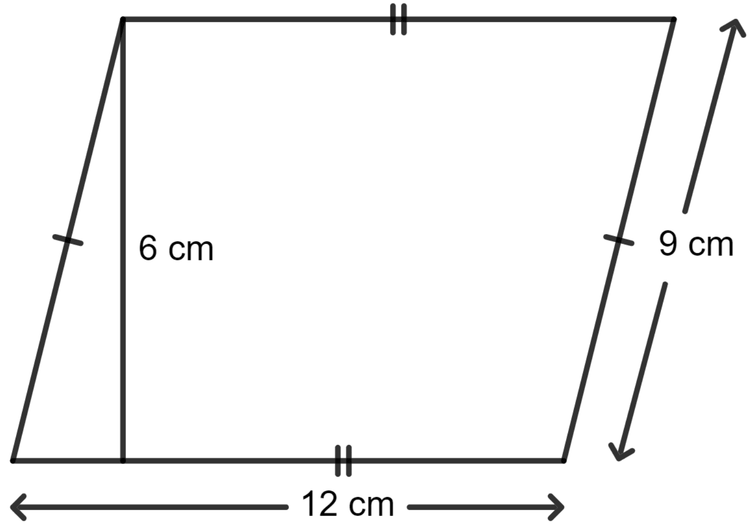
1. 12 cm
2. 9 cm
3. 8 cm
4. 6 cm

Answer

Given:

The base of the parallelogram = 12 cm.

The height of the parallelogram = 6 cm.



As we know, the area of parallelogram = base x height

The area of the parallelogram using the longer side:

= 12 x 6 cm^2

= 72 cm^2.

The base of the parallelogram = 9 cm.

Let the height of the parallelogram be b cm.

The area of the parallelogram using the shorter side:

= 9 x b cm^2

Since the area is the same, we can equate both areas:

⇒ 9 x b cm^2 = 72 cm^2

⇒ b = $\dfrac{72}{9}$ cm

⇒ b = 8 cm

Hence, option 3 is the correct option.

Question 1(iv)

Each side of a rhombus is 9 cm and the distance between its opposite sides is 6 cm. The area of the rhombus is :

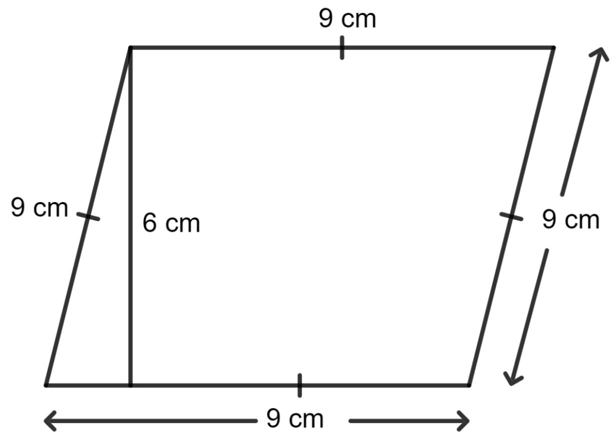
1. 54 cm^2
2. 108 cm^2
3. 27 cm^2
4. 13.5 cm^2

Answer

Given:

The base of the rhombus = 9 cm.

The height of the rhombus = 6 cm.



As we know, the area of a rhombus = base x height

= 9 x 6 cm^2

= 54 cm^2

Hence, option 1 is the correct option.

Question 1(v)

One diagonal of a rhombus is 15 cm and its area is 60 cm^2; the other diagonal of the rhombus is :

1. 4 cm
2. 8 cm
3. 12 cm
4. 16 cm

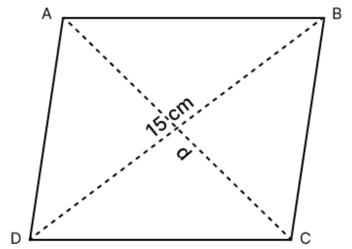
Answer

Given:

One diagonal of a rhombus = 15 cm.

Area = 60 cm^2

Let the other diagonal of the rhombus be `d`.



As we know, the area of a rhombus = $\dfrac{1}{2}$ x product of its diagonals

⇒ $\dfrac{1}{2}$ x (15 x d) = 60

⇒ $\dfrac{15}{2}$ x d = 60

⇒ d = $\dfrac{60 \times 2}{15}$ cm

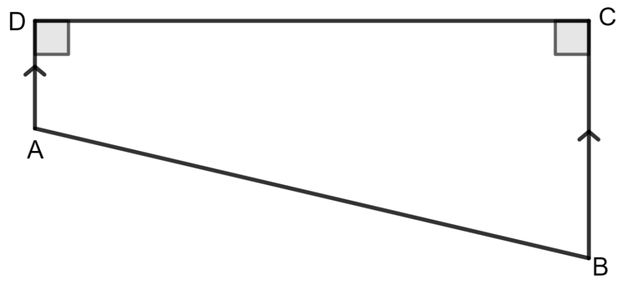
⇒ d = $\dfrac{120}{15}$ cm

⇒ d = 8 cm

Hence, option 2 is the correct option.

Question 2

The following figure shows the cross-section ABCD of a swimming pool which is a trapezium in shape.



If the width DC of the swimming pool is 6.4 m, depth (AD) at the shallow end is 80 cm and depth (BC) at the deepest end is 2.4 m, find its area of cross-section.

Answer

Given:

DC = 6.4 m

AD = 80 cm = $\frac{80}{100}$ m = 0.8 m

BC = 2.4 m

Area of the cross-section = Area of trapezium ABCD

= $\frac{1}{2}$ x (sum of parallel sides) x height

= $\frac{1}{2}$ x (0.8 + 2.4) x 6.4

= $\frac{1}{2}$ x 3.2 x 6.4

= $\frac{1}{2}$ x 20.48

= 10.24 m^2

Hence, the area of cross-section is 10.24 m^2.

Question 3

The parallel sides of a trapezium are in the ratio 3 : 4. If the distance between the parallel sides is 9 dm and its area is 126 dm^2, find the lengths of its parallel sides.

Answer

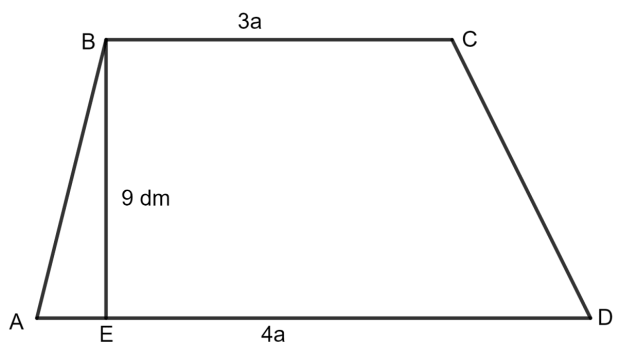
Given:

The parallel sides of a trapezium are in the ratio 3 : 4.

The distance between the parallel sides = 9 dm

The area = 126 dm^2

Let the parallel sides of trapezium be 3a and 4a.



As we know, the area of a trapezium = $\frac{1}{2}$ x (sum of parallel side) x height

⇒ $\frac{1}{2}$ x (3a + 4a) x 9 = 126

⇒ $\frac{1}{2}$ x 7a x 9 = 126

⇒ $\frac{1}{2}$ x 63a = 126

⇒ 63a = 126 x 2

⇒ 63a = 252

⇒ a = $\frac{252}{63}$

⇒ a = 4

So, the parallel sides of the trapezium are:

For 3a:

= 3 x 4 dm = 12 dm

And, for 4a:

= 4 x 4 dm = 16 dm

Hence, the lengths of its parallel sides are 12 dm and 16 dm.

Question 4

The two parallel sides and the distance between them are in the ratio 3 : 4 : 2. If the area of the trapezium is 175 cm^2, find its height.

Answer

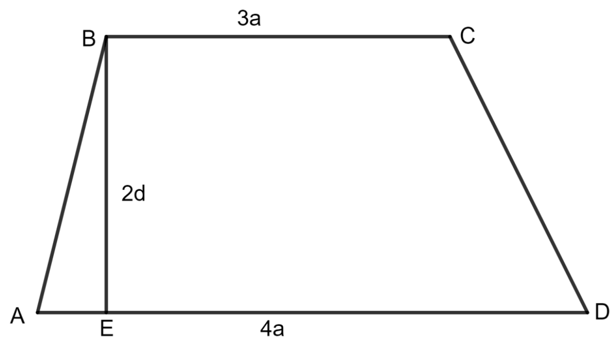
Given:

The two parallel sides and the distance between them are in the ratio 3 : 4 : 2.

The area of the trapezium is 175 cm^2.

So, the two parallel sides of the trapezium are 3a and 4a.

Distance between them is 2a.



As we know, the area of a trapezium = $\frac{1}{2}$ x (sum of parallel side) x height

⇒ $\frac{1}{2}$ x (3a + 4a) x 2a = 175

⇒ $\frac{1}{2}$ x 7a x 2a = 175

⇒ $\frac{1}{2}$ x $14a^2$ = 175

⇒ $7a^2$ = 175

⇒ $a^2 = \frac{175}{7}$

⇒ $a^2 = 25$

⇒ a = $\sqrt{25}$

⇒ a = 5

So, height of the trapezium = 2a

= 2 x 5 cm

= 10 cm

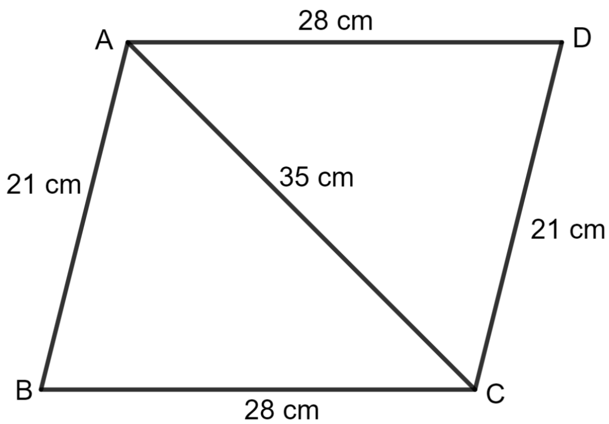
Hence, the height of the trapezium is 10 cm.

Question 5

The adjacent sides of a parallelogram are 21 cm and 28 cm. If its one diagonal is 35 cm, find the area of the parallelogram.

Answer

Let the parallelogram be ABCD as shown in the figure below:



For triangle ABC,

AB = 21 cm, BC = 28 cm and AC = 35 cm.

$\because s = \dfrac{a + b + c}{2}$

$= \dfrac{21 + 28 + 35}{2}$

$= \dfrac{84}{2}$

$= 42$

∵ Area of triangle ABC = $\sqrt{s(s - a)(s - b)(s - c)}$

= $\sqrt{42(42 - 21)(42 - 28)(42 - 35)}$ cm^2

= $\sqrt{42 \times 21 \times 14 \times 7}$ cm^2

= $\sqrt{86,436}$ cm^2

= 294 cm^2

∵ Diagonal of parallelogram divides it into two equal parts.

So, area of parallelogram ABCD = 2 x area of triangle ABC.

= 2 x 294 cm^2

= 588 cm^2

Hence, the area of the parallelogram is 588 cm^2.

Question 6

The diagonals of a rhombus are 18 cm and 24 cm. Find :

(i) its area

(ii) length of its sides

(iii) its perimeter

Answer

(i) **Given:**

The diagonals of a rhombus are 18 cm and 24 cm.

As we know, the area of rhombus = $\frac{1}{2}$ x product of its diagonal

= $\frac{1}{2}$ x 18 x 24 cm^2

= $\frac{1}{2}$ x 432 cm^2

= 216 cm^2

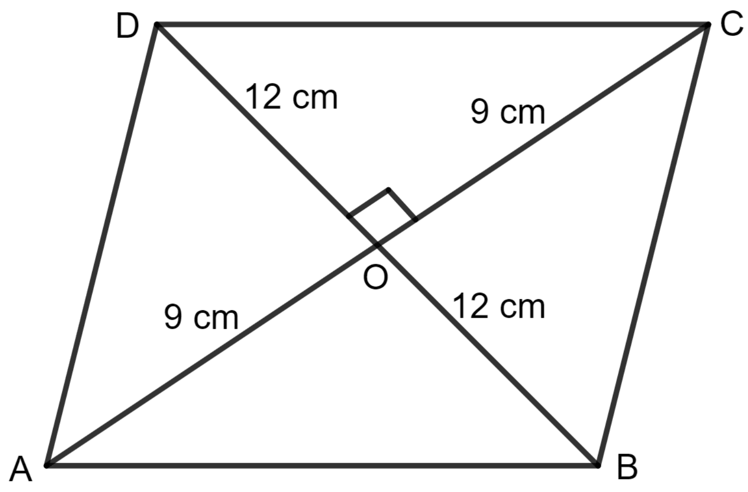
Hence, the area of the rhombus is 216 cm^2.

(ii) AC = 18 cm

Then, OA = OC = $\frac{18}{2}$ = 9 cm

And, BD = 24 cm

Then, OB = OD = $\frac{24}{2}$ = 12 cm



Since the diagonals of a rhombus bisect at 90°.

Applying pythagoras theorem in triangle AOB, we get:

$AB^2 = OA^2 + OB^2$

$\Rightarrow AB^2 = (9)^2 + (12)^2$

$\Rightarrow AB^2 = 81 + 144$

$\Rightarrow AB^2 = 225$

$\Rightarrow AB = \sqrt{225}$

$\Rightarrow AB = 15$ cm

Hence, the length of each side of the rhombus is 15 cm.

(iii) As we know, the perimeter of the rhombus = 4 x side

= 4 x 15 cm

= 60 cm

Hence, the perimeter of the rhombus is 60 cm.

Question 7

The perimeter of a rhombus is 40 cm. If one diagonal is 16 cm, find :

(i) its other diagonal

(ii) its area

Answer

(i) **Given:**

The perimeter of a rhombus = 40 cm.

One diagonal = 16 cm.

As we know, the perimeter of the rhombus = 4 x side

⇒ 4 x side = 40 cm

⇒ Side = $\dfrac{40}{4}$ cm

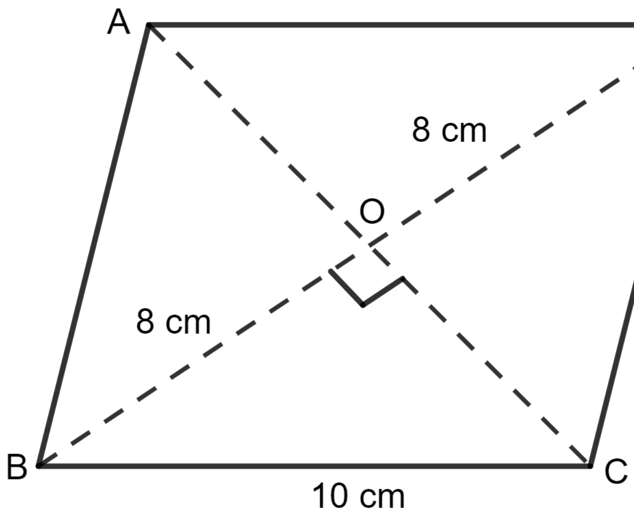
⇒ Side = 10 cm

AB = 10 cm

AC = 16 cm

Then, OA = OC = $\dfrac{16}{2}$ = 8 cm

Let OB be a cm.



Since the diagonal of a rhombus bisect at 90°.

Applying pythagoras theorem in triangle AOB, we get:

$AB^2 = OA^2 + OB^2$

$\Rightarrow (10)^2 = (8)^2 + a^2$

$\Rightarrow 100 = 64 + a^2$

$\Rightarrow a^2 = 100 - 64$

$\Rightarrow a^2 = 36$

$\Rightarrow a = \sqrt{36}$

$\Rightarrow a = 6$

So, OB = 6 cm

BD = 2 x 6 cm

= 12 cm

Hence, the length of other diagonal is 12 cm.

(ii) As we know that area of rhombus = $\frac{1}{2}$ x product of its diagonal

= $\frac{1}{2}$ x 16 x 12 cm^2

= $\frac{1}{2}$ x 192 cm^2

= 96 cm^2

Hence, the area of the rhombus is 96 cm^2.

Question 8

The length of the diagonals of a rhombus is in the ratio 4 : 3. If its area is 384 cm^2, find its side.

Answer

Given:

The length of the diagonals of a rhombus is in the ratio 4 : 3.

The area = 384 cm^2

Let the length of the diagonals be 4a and 3a.

As we know, the area of rhombus = $\frac{1}{2}$ x product of its diagonal

$\Rightarrow \dfrac{1}{2}$ x 4a x 3a = 384

$\Rightarrow \dfrac{1}{2}$ x 12a^2 = 384

$\Rightarrow$ 6a^2 = 384

$\Rightarrow$ a^2 = $\dfrac{384}{6}$

$\Rightarrow$ a^2 = 64

$\Rightarrow$ a = $\sqrt{64}$

$\Rightarrow$ a = 8

The length of the diagonals be 4a and 3a.

= 4 x 8 cm and 3 x 8 cm

= 32 cm and 24 cm

AC = 32 cm

Then, OA = OC = $\dfrac{32}{2}$ = 16 cm

And, BD = 24 cm

Then, OB = OD = $\dfrac{24}{2}$ = 12 cm

Since the diagonal of a rhombus bisect at 90°.

Applying pythagoras theorem in ΔAOB, we get:

$AB^2 = OA^2 + OB^2$

$\Rightarrow AB^2 = (16)^2 + (12)^2$

$\Rightarrow AB^2 = 256 + 144$

$\Rightarrow AB^2 = 400$

$\Rightarrow AB = \sqrt{400}$

$\Rightarrow$ AB = 20 cm

Hence, the length of its side is 20 cm.

Question 9

A thin metal iron-sheet is a rhombus in shape, with each side 10 m. If one of its diagonals is 16 m, find the cost of painting its both sides at the rate of ₹ 6 per m^2.

Also, find the distance between the opposite sides of this rhombus.

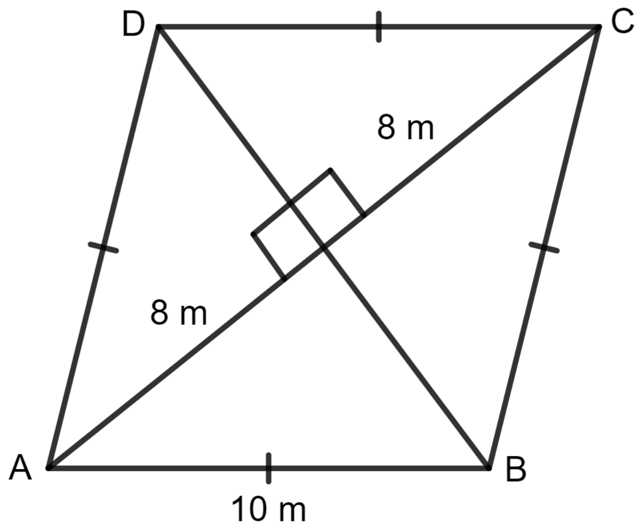
Answer

Given:

Side of iron sheet (AB) = 10 m

Diagonal of the sheet (AC) = 16 m

Join BD.



Then, OA = OC = $\dfrac{AC}{2}$ = $\dfrac{16}{2}$ = 8 m

Since the diagonals of a rhombus bisect each other at 90°, applying the pythagoras theorem in Δ AOB, we get:

$AB^2 = OA^2 + OB^2$

$\Rightarrow (10)^2 = (8)^2 + OB^2$

$\Rightarrow 100 = 64 + OB^2$

$\Rightarrow OB^2 = 100 - 64$

$\Rightarrow OB^2 = 36$

$\Rightarrow OB = \sqrt{36}$

$\Rightarrow$ OB = 6 m

Thus, BD = 2 x OB = 2 x 6 m = 12 m

The area of rhombus = $\frac{1}{2}$ x product of diagonals

= $\frac{1}{2}$ x 16 x 12 m^2

= $\frac{1}{2}$ x 192 m^2

= 96 m^2

The rate of painting is ₹ 6 per m^2.

Therefore, the total cost of painting is:

Total cost = 2 x area of sheet x rate of painting

= 2 x 96 x 6

= 192 x 6

= ₹ 1,152

We also know that the area of the rhombus = base x height

⇒ 96 = 10 x height

⇒ height = $\frac{96}{10}$

⇒ height = 9.6 m

Hence, the total cost of painting is ₹ 1,152 and the distance between the opposite sides of the rhombus is 9.6 m.

Question 10

The area of a trapezium is 279 sq. cm and the distance between its two parallel sides is 18 cm. If one of its parallel sides is longer than the other side by 5 cm, find the lengths of its parallel sides.

Answer

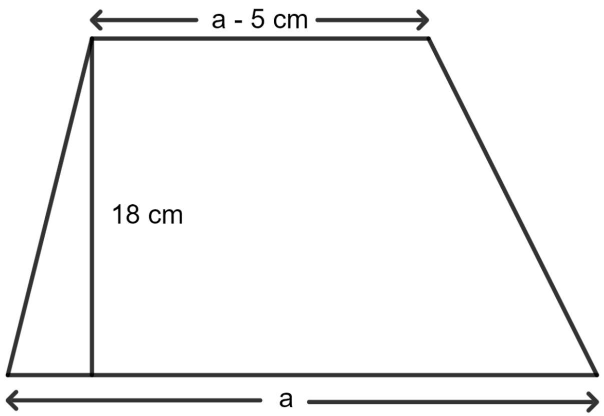
Given:

The area of a trapezium = 279 sq. cm

The distance between its two parallel sides = 18 cm.

One of the parallel sides is longer than the other side by 5 cm.

Let one of the parallel sides be a. Then, the other parallel side is a - 5.



As we know, the area of a trapezium = $\frac{1}{2}$ (sum of parallel sides) x distance between the parallel sides

Substituting the given values:

⇒ 279 = $\frac{1}{2}$ (a + a - 5) x 18

⇒ 279 = $\frac{1}{2}$ (2a - 5) x 18

⇒ 279 = $\frac{1}{2}$ (36a - 90)

⇒ 279 = 18a - 45

⇒ 18a = 279 + 45

⇒ 18a = 324

⇒ a = $\frac{324}{18}$

⇒ a = 18 cm

So, the parallel sides are 18 cm and (18 - 5) = 13 cm.

Hence, the lengths of the parallel sides are 18 cm and 13 cm, respectively.

Question 11

The area of a rhombus is equal to the area of a triangle. If base of triangle is 24 cm, its corresponding altitude is 16 cm and one of the diagonals of the rhombus is 19.2 cm, find its other diagonal.

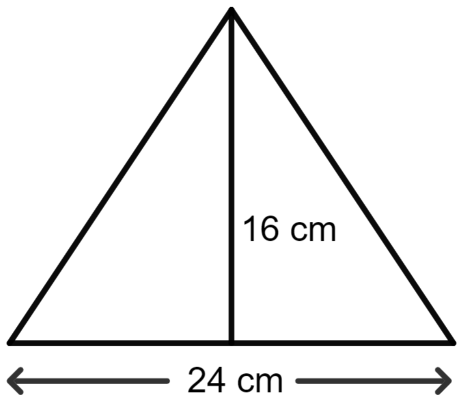
Answer

Given:

The area of a rhombus = The area of a triangle

The base of the triangle = 24 cm

The altitude of the triangle = 16 cm



One of the diagonals of the rhombus = 19.2 cm

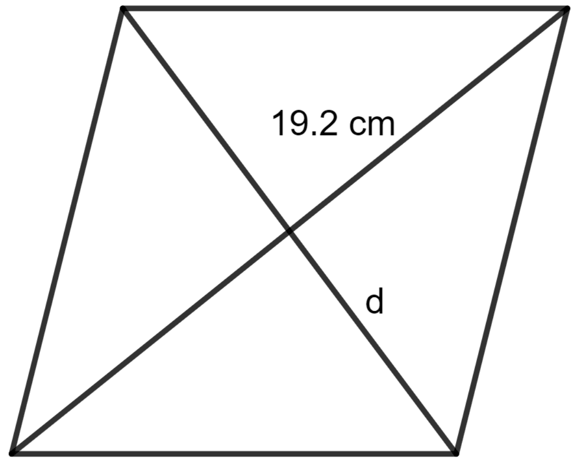
As we know, the area of a triangle = $\frac{1}{2}$ x base x altitude

= $\frac{1}{2}$ x 24 x 16 cm^2

= $\frac{1}{2}$ x 384 cm^2

= 192 cm^2

Let the length of the other diagonal of the rhombus be `d`.



The area of the rhombus = $\frac{1}{2}$ x (product of diagonals)

⇒ $\frac{1}{2}$ x (19.2 x d) = 192

⇒ 9.6 x d = 192

⇒ d = $\frac{192}{9.6}$

⇒ d = 20 cm

Hence, the length of the other diagonal is 20 cm.

Question 12

Find the area of the trapezium ABCD in which AB//DC, AB = 18 cm, ∠B = ∠C = 90°, CD = 12 cm and AD = 10 cm.

Answer

Given:

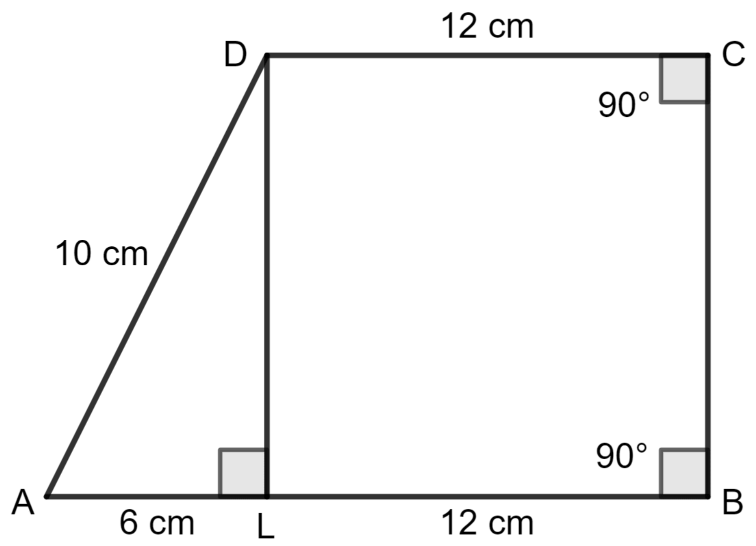
AB = 18 cm

$\angle B = \angle C = 90°$

CD = 12 cm

AD = 10 cm

Draw DL perpendicular to AB.



Since, AL = AB - DC

= 18 - 12

= 6 cm

Also, DL = BC (Both are perpendiculars)

Using pythagoras theorem,

$AD^2 = AL^2 + LD^2$

⇒ $10^2 = 6^2 + LD^2$

⇒ $100 = 36 + LD^2$

⇒ $LD^2 = 100 - 36$

⇒ $LD^2 = 64$

⇒ LD = $\sqrt{64}$

⇒ LD = 8 cm

As we know, the area of a trapezium = $\frac{1}{2}$ (sum of parallel sides) x distance betweem the parallel sides.

= $\frac{1}{2}$ (18 + 12) x 8

= $\frac{1}{2}$ x 30 x 8

= $\frac{1}{2}$ x 240

= 120 cm^2

Hence, the area of the trapezium is 120 cm^2.

Exercise 22(D)

Question 1(i)

The circumference of a circle is numerically same as its area; then its radius is :

1. 4 unit
2. 2 unit
3. 1 unit
4. none of these

Answer

Given:

The circumference of a circle = The area of a circle

Let r be the radius of the circle.

As we know, the circumference of the circle = 2πr

And, the area of the circle = πr^2

So,

⇒ $2\pi r = \pi r^2$

⇒ $2r = r^2$

⇒ $2 = r$

Hence, the radius of the circle is 2 units.

Hence, option 2 is the correct option.

Question 1(ii)

The perimeter of a square of side 11 cm is equal to circumference of a circle; then the diameter of the circle is :

1. 7 cm
2. 7.5 cm
3. 14 cm
4. 3.5 cm

Answer

Given:

The perimeter of a square = The circumference of a circle

Side of the square = 11 cm

As we know, the perimeter of a square = 4 x side

= 4 x 11 cm

= 44 cm

Let `r` be the radius of the circle.

And, the circumference of the circle = 2πr

⇒ 2πr = 44

Substitute π = $\dfrac{22}{7}$:

$$\Rightarrow 2 \times \dfrac{22}{7} \times r = 44$$

$$\Rightarrow \dfrac{44}{7} \times r = 44$$

$$\Rightarrow r = \dfrac{44 \times 7}{44}$$

$$\Rightarrow r = \dfrac{\cancel{44} \times 7}{\cancel{44}}$$

$$\Rightarrow r = 7 \text{ cm}$$

So, the diameter of the circle = 2 x radius

= 2 x 7 cm

= 14 cm

Hence, option 1 is the correct option.

Question 1(iii)

The circumference of a circle is equal to the sum of the circumferences of two circles with radii 10 cm and 12 cm. The radius of the circle formed is :

1. 2 cm
2. 44 cm
3. 22 cm
4. 244 cm

Answer

Given:

The circumference of a circle = The sum of the circumferences of two circles with radii 10 cm and 12 cm.

As we know, the circumference of a circle = 2πr

For r = 10 cm:

Circumference of the circle = 2 x $\dfrac{22}{7}$ x 10

= $\dfrac{44}{7}$ x 10

= $\dfrac{440}{7}$

For r = 12 cm

Circumference of the circle = 2 x $\dfrac{22}{7}$ x 12

= $\dfrac{44}{7}$ x 12

= $\dfrac{528}{7}$

Let R be the radius of the larger circle.

The circumference of the larger circle =

$\dfrac{440}{7} + \dfrac{528}{7}$

$$\Rightarrow 2\pi R = \frac{440 + 528}{7}$$

$$\Rightarrow 2 \times \frac{22}{7} \times R = \frac{440 + 528}{7}$$

$$\Rightarrow \frac{44}{7} \times R = \frac{968}{7}$$

$$\Rightarrow R = \frac{968 \times 7}{7 \times 44}$$

$$\Rightarrow R = \frac{968 \times \cancel{7}}{\cancel{7} \times 44}$$

$$\Rightarrow R = \frac{968}{44}$$

$$\Rightarrow R = 22 \text{ cm}$$

Hence, option 3 is the correct option.

Question 1(iv)

The diameter of a circular wheel is 56 cm. The distance moved by it in 100 rounds is :

1. 176 cm
2. 17600 cm
3. 1760 cm
4. 352 cm

Answer

Given:

The diameter of a circular wheel = 56 cm

Total number of rounds = 100

Since diameter = 2 x Radius

⇒ Radius = $\dfrac{\text{Diameter}}{2}$

⇒ Radius = $\dfrac{56}{2}$

= 28 cm

As we know, the circumference of a circle = 2πr

$$= 2 \times \frac{22}{7} \times 28$$

$$= \frac{44}{7} \times 28$$

$$= \frac{1,232}{7}$$

$$= 176 \text{ cm}$$

Total distance = circumference of the circle x number of rounds

= 176 x 100

= 17600 cm

Hence, option 2 is the correct option.

Question 1(v)

A circular wheel of radius 3.5 cm makes 600 rounds in 48 seconds, its speed is :

1. 275 cm/s
2. 27.5 m/s
3. 550 cm/s
4. 55 cm/s

Answer

Given:

Radius of circular wheel = 3.5 cm

Total rounds = 600

Total time = 48 seconds

As we know, the circumference of a circle = 2πr

$= 2 \times \dfrac{22}{7} \times 3.5$

$= \dfrac{44}{7} \times 3.5$

$= \dfrac{154}{7}$

$= 22 \text{ cm}$

Total distance = The circumference of the circle x number of rounds

= 22 x 600

= 13200 cm

And, Speed = $\dfrac{\text{Distance}}{\text{Time}}$

= $\dfrac{13200}{48}$

= 275 cm/s

Hence, option 1 is the correct option.

Question 2

Find the radius and area of a circle, whose circumference is :

(i) 132 cm

(ii) 22 m

Answer

(i) Let `r` be the radius of the circle.

The circumference of a circle = 132 cm

As we know, the circumference of the circle = 2πr

$$\Rightarrow 2 \times \dfrac{22}{7} \times r = 132$$

$$\Rightarrow \dfrac{44}{7} \times r = 132$$

$$\Rightarrow r = \dfrac{132 \times 7}{44}$$

$$\Rightarrow r = \dfrac{924}{44}$$

$$\Rightarrow r = 21 \text{ cm}$$

And, area of the circle = πr^2

$$= \dfrac{22}{7} \times 21^2$$

$$= \dfrac{22}{7} \times 441$$

$$= \dfrac{9,702}{7}$$

$$= 1,386 \text{ cm}^2$$

Hence, the radius of the circle is 21 cm and the area is 1,386 cm^2.

(ii) Let `r` be the radius of the circle.

The circumference of a circle = 22 cm

As we know, the circumference of the circle = 2πr

$$\Rightarrow 2 \times \frac{22}{7} \times r = 22$$

$$\Rightarrow \frac{44}{7} \times r = 22$$

$$\Rightarrow r = \frac{22 \times 7}{44}$$

$$\Rightarrow r = \frac{154}{44}$$

$$\Rightarrow r = 3.5 \text{ m}$$

And, area of the circle = πr^2

$$= \frac{22}{7} \times 3.5^2$$

$$= \frac{22}{7} \times 12.25$$

$$= \frac{269.50}{7}$$

$$= 38.5 \text{ m}^2$$

Hence, the radius of the circle is 3.5 m and the area is 38.5 m^2.

Question 3

Find the radius and circumference of a circle, whose area is :

(i) 154 cm^2

(ii) 6.16 m^2

Answer

(i) Let `r` be the radius of the circle.

The area of circle = 154 cm^2

As we know, the area of the circle = πr^2

$$\Rightarrow \frac{22}{7} \times r^2 = 154$$

$$\Rightarrow r^2 = \frac{154 \times 7}{22}$$

$$\Rightarrow r^2 = \frac{1,078}{22}$$

$$\Rightarrow r^2 = 49$$

$$\Rightarrow r = \sqrt{49}$$

$$\Rightarrow r = 7 \text{ cm}$$

Circumference of the circle = $2\pi r$

$$= 2 \times \frac{22}{7} \times 7$$

$$= \frac{44}{7} \times 7$$

$$= \frac{308}{7}$$

$$= 44 \text{ cm}$$

Hence, the radius of the circle is 7 cm and the circumference is 44 cm.

(ii) Let r be the radius of the circle.

The area of circle = 6.16 m^2

As we know, the area of the circle = πr^2

$$\Rightarrow \frac{22}{7} \times r^2 = 6.16$$

$$\Rightarrow r^2 = \frac{6.16 \times 7}{22}$$

$$\Rightarrow r^2 = \frac{43.12}{22}$$

$$\Rightarrow r^2 = 1.96$$

$$\Rightarrow r = \sqrt{1.96}$$

$$\Rightarrow r = 1.4 \text{ m}$$

Circumference of the circle = $2\pi r$

$$= 2 \times \frac{22}{7} \times 1.4$$

$$= \frac{44}{7} \times 1.4$$

$$= \frac{61.6}{7}$$

$$= 8.8 \text{ m}$$

Hence, the radius of the circle is 1.4 m and the circumference is 8.8 m.

Question 4

The circumference of a circular table is 88 m. Find its area.

Answer

Given:

The circumference of a circular table = 88 m

Let r be the radius of the circle.

As we know, the circumference of the circle = 2πr

$$\Rightarrow 2 \times \frac{22}{7} \times r = 88$$

$$\Rightarrow \frac{44}{7} \times r = 88$$

$$\Rightarrow r = \frac{88 \times 7}{44}$$

$$\Rightarrow r = \frac{616}{44}$$

$$\Rightarrow r = 14 \text{ m}$$

The area of the circle = πr^2

$$= \frac{22}{7} \times 14^2$$

$$= \frac{22}{7} \times 196$$

$$= \frac{4,312}{7}$$

$$= 616 \text{ m}^2$$

Hence, the area of the circular table is 616 m^2.

Question 5

The area of a circle is 1386 sq. cm, find its circumference.

Answer

Given:

The area of a circle = 1386 sq. cm

Let `r` be the radius of the circle.

As we know, the area of the circle = πr^2

$$\Rightarrow \frac{22}{7} \times r^2 = 1386$$

$$\Rightarrow r^2 = \frac{1386 \times 7}{22}$$

$$\Rightarrow r^2 = \frac{9,702}{22}$$

$$\Rightarrow r^2 = 441$$

$$\Rightarrow r = \sqrt{441}$$

$$\Rightarrow r = 21 \text{ cm}$$

Circumference of the circle = 2πr

$$= 2 \times \frac{22}{7} \times 21$$

$$= \frac{44}{7} \times 21$$

$$= \frac{324}{7}$$

$$= 132 \text{ cm}$$

Hence, the circumference of the circle is 132 cm.

Question 6

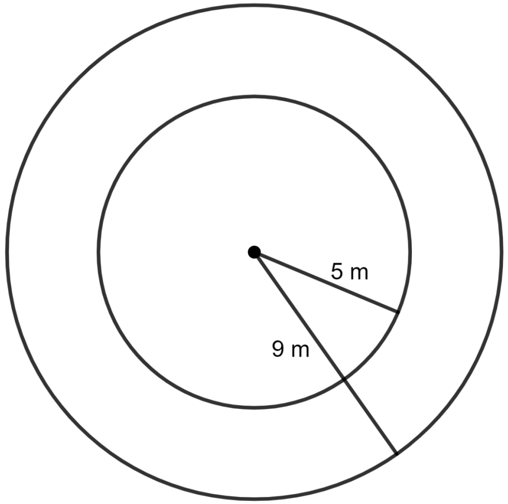
Find the area of a flat circular ring formed by two concentric circles (circles with same centre) whose radii are 9 cm and 5 cm.

Answer

Given:

Radius of outer circle = 9 cm

Radius of inner circle = 5 cm



Area of circular ring = Area of outer circle - Area of inner circle

As we know, the area of the circle = πr^2

Area of circular ring =

$$\frac{22}{7} \times 9^2 - \frac{22}{7} \times 5^2$$

$$= \frac{22}{7} \times 81 - \frac{22}{7} \times 25$$

$$= \frac{1,782}{7} - \frac{550}{7}$$

$$= \frac{1,782 - 550}{7}$$

$$= \frac{1,232}{7}$$

$$= 176 \text{ cm}^2$$

Hence, the area of the circular ring is 176 cm^2.

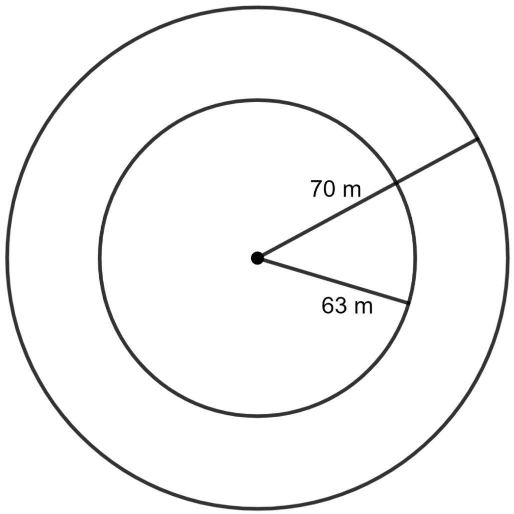
Question 7

The radii of the inner and outer circumferences of a circular-running-track are 63 m and 70 m respectively. Find :

(i) the area of the track

(ii) the difference between the lengths of the two circumferences of the track

Answer



(i) **Given:**

Radius of outer circle = 70 m

Radius of inner circle = 63 m

Area of circular track = Area of outer circle - Area of inner circle

As we know, the area of the circle = πr^2

Area of circular ring =

$$\frac{22}{7} \times 70^2 - \frac{22}{7} \times 63^2$$

$$= \frac{22}{7} \times 4,900 - \frac{22}{7} \times 3,969$$

$$= \frac{1,07,800}{7} - \frac{87,318}{7}$$

$$= \frac{1,07,800 - 87,318}{7}$$

$$= \frac{20,482}{7}$$

$$= 2,926 \text{ m}^2$$

Hence, the area of the circular track is 2,926 m^2.

(ii) The difference between the lengths of the two circumferences of the track = Circumference of outer circle - Circumference of inner circle

As we know, the circumference of the circle = $2\pi r$

The difference between the lengths of the two circumferences of the track =

$$2 \times \frac{22}{7} \times 70 - 2 \times \frac{22}{7} \times 63$$

$$= \frac{44}{7} \times 70 - \frac{44}{7} \times 63$$

$$= \frac{3080}{7} - \frac{2772}{7}$$

$$= \frac{3080 - 2772}{7}$$

$$= \frac{308}{7}$$

$$= 44 \text{ m}$$

Hence, the difference between the lengths of the two circumferences of the track is 44 m.

Question 8

A circular field of radius 105 m has a circular path of uniform width of 5 m along and inside its boundary. Find the area of the path.

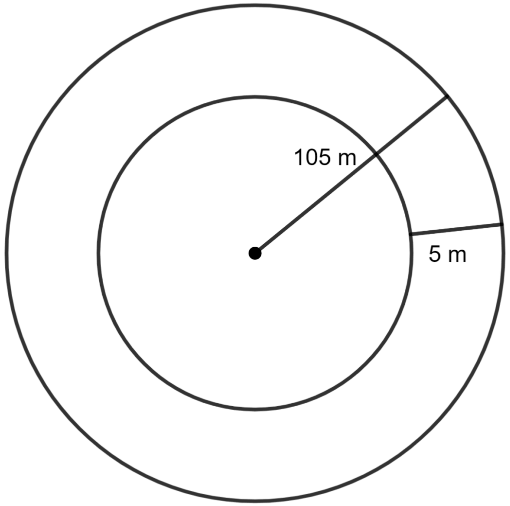
Answer

Given:

Radius of outer circular field = 105 m

Radius of inner circular field = 105 - 5 m

= 100 m



Area of circular ring = Area of outer circular field - Area of inner circular field

As we know, the area of the circle = πr^2

Area of circular ring =

$$\dfrac{22}{7} \times 105^2 - \dfrac{22}{7} \times 100^2$$

$$= \dfrac{22}{7} \times 11,025 - \dfrac{22}{7} \times 10,000$$

$$= \dfrac{2,42,500}{7} - \dfrac{2,20,000}{7}$$

$$= \dfrac{2,42,500 - 2,20,000}{7}$$

$$= \dfrac{22,550}{7}$$

$$= 3221\dfrac{3}{7} \text{ m}^2$$

Hence, the area of circular field is $3221\dfrac{3}{7}$ m^2.

Question 9

A wire, when bent in the form of a square, encloses an area of 484 cm^2. Find :

(i) one side of the square

(ii) length of the wire

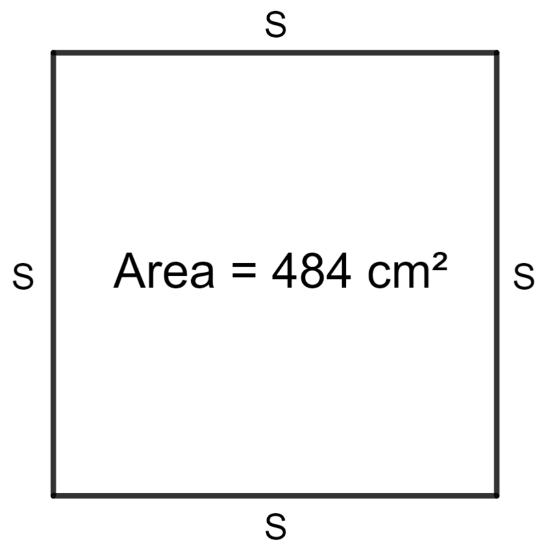
(iii) the largest area enclosed, if the same wire is bent to form a circle.

Answer

Given:

Area of the square = 484 cm^2

Let s be the side of the square.



As we know, the area of the square = side^2

⇒ s^2 = 484

⇒ s = $\sqrt{484}$

⇒ s = 22

Hence, one side of square is 22 cm.

(ii) Total length of the wire = perimeter of the square

As we know, the perimeter of the square = 4 x side

= 4 x 22

= 88 cm

Hence, the length of the wire is 88 cm.

(iii) Perimeter of the square = circumference of the circle

Let `r` be the radius of the circle.

⇒ 2πr = 88 cm

$$\Rightarrow 2 \times \frac{22}{7} \times r = 88$$

$$\Rightarrow \frac{44}{7} \times r = 88$$

$$\Rightarrow r = \frac{7 \times 88}{44}$$

$$\Rightarrow r = \frac{616}{44}$$

$$\Rightarrow r = 14 \text{ cm}$$

Area of the circle = πr^2

$$= \frac{22}{7} \times 14^2$$

$$= \frac{22}{7} \times 196$$

$$= \frac{4312}{7}$$

$$= 616 \text{ cm}^2$$

Hence, the largest area that can be enclosed when the same wire is bent to form a circle is 616 cm^2.

Question 10

A wire, when bent in the form of a square, encloses an area of 196 cm^2. If the same wire is bent to form a circle, find the area of the circle.

Answer

Given:

Area of the square = 196 cm^2

Let s be the side of the square.

As we know, the area of the square = side^2

⇒ $s^2 = 196$

⇒ s = $\sqrt{196}$

⇒ s = 14

As we know, the perimeter of the square = 4 x side

= 4 x 14

= 56 cm

Perimeter of the square = circumference of the circle

Let `r` be the radius of the circle.

⇒ 2πr = 56

$$\Rightarrow 2 \times \frac{22}{7} \times r = 56$$

$$\Rightarrow \frac{44}{7} \times r = 56$$

$$\Rightarrow r = \frac{7 \times 56}{44}$$

$$\Rightarrow r = \frac{392}{44}$$

$$\Rightarrow r = 8\frac{10}{11} \text{ cm}$$

Area of the circle = πr^2

$$= \frac{22}{7} \times \left(8\frac{10}{11}\right)^2$$

$$= \frac{22}{7} \times \left(\frac{98}{11}\right)^2$$

$$= \frac{22}{7} \times \left(\frac{9604}{121}\right)$$

$$= \frac{211,288}{847}$$

$$= 249.45 \text{ cm}^2$$

Hence, the area of the circle is 249.45 cm^2.

Question 11

The radius of a circular wheel is 42 cm. Find the distance traveled by it in :

(i) 1 revolution

(ii) 50 revolutions

(iii) 200 revolutions

Answer

(i) **Given:**

The radius of a circular wheel = 42 cm = 0.42 m

As we know, the circumference of the circle = 2πr

$$2 \times \frac{22}{7} \times 0.42$$

$$= \frac{44}{7} \times 0.42$$

$$= \frac{18.48}{7}$$

$$= 2.64 \text{ m}$$

Hence, the distance traveled by wheel in 1 revolution is 2.64 m.

(ii) Distance traveled by wheel in 50 revolutions = Circumference of the wheel x 50

= 2.64 x 50 m

= 132 m

Hence, the distance traveled by wheel in 50 revolutions is 132 m.

(ii) Distance traveled by wheel in 200 revolutions = Circumference of the wheel x 200

= 2.64 x 200 m

= 528 m

Hence, the distance traveled by wheel in 200 revolutions is 528 m.

Question 12

The diameter of wheel is 0.70 m. Find the distance covered by it in 500 revolutions.

If the wheel takes 5 minutes to make 500 revolutions; find its speed in :

(i) m/s

(ii) km/hr

Answer

(i)**Given:**

Diameter of wheel = 0.70 m

Radius = $\dfrac{\text{Diameter}}{2}$

= $\dfrac{0.70}{2}$ m

= 0.35 m

As we know, the circumference of the circle = 2πr

$$2 \times \frac{22}{7} \times 0.35$$

$$= \frac{44}{7} \times 0.35$$

$$= \frac{15.40}{7}$$

$$= 2.2 \text{ m}$$

Distance traveled by wheel in 500 revolutions = Circumference of the wheel x 500

= 2.2 x 500 m

= 1100 m

Speed = $\dfrac{\text{Distance}}{\text{Time}}$

Time taken by wheel = 5 min = 5 x 60 sec = 300 sec.

$$\text{Speed} = \dfrac{1100}{300}$$

$$= 3\dfrac{2}{3} \text{ m/s}$$

Hence, the speed = $3\dfrac{2}{3}$ m/s.

(ii) Speed = $3\dfrac{2}{3}$ m/s

$$= 3\dfrac{2}{3} \times \dfrac{3600}{1000} \text{ km/hr}$$

$$= \dfrac{11}{3} \times \dfrac{3600}{1000} \text{ km/hr}$$

$$= \dfrac{39600}{3000} \text{ km/hr}$$

$$= 13.2 \text{ km/hr}$$

Hence, the speed = 13.2 km/hr.

Question 13

A bicycle wheel, diameter 56 cm, is making 45 revolutions in every 10 seconds.
At what speed, in kilometre per hour, is the bicycle traveling ?

Answer

Given:

Diameter of wheel = 56 cm

Diameter of wheel in Kilometre = 0.00056 km

Radius = $\dfrac{\text{Diameter}}{2}$

= $\dfrac{0.00056}{2}$

= 0.00028 km

As we know, the circumference of the circle = 2πr

$= 2 \times \dfrac{22}{7} \times 0.00028$

$= \dfrac{44}{7} \times 0.00028$

$= \dfrac{0.01232}{7}$

$= 0.00176 \text{ km}$

Distance traveled by wheel in 45 revolutions = Circumference of the wheel x 45

= 0.00176 x 45 km

= 0.0792 km

And, speed = $\dfrac{\text{Distance}}{\text{Time}}$

It is given that time taken by wheel = 10 sec = $\dfrac{10}{3600}$.

$$\text{Speed} = \frac{0.0792 \times 3600}{10}$$

$$= \frac{285.12}{10}$$

$$= 28.512 \text{ km/hr}$$

Hence, the speed = 28.512 km/hr.

Question 14

A roller has a diameter of 1.4 m. Find :

(i) its circumference

(ii) the number of revolutions it makes while traveling 61.6 m.

Answer

(i) **Given:**

Diameter of roller = 1.4 m

Radius = $\frac{\text{Diameter}}{2}$

= $\frac{1.4}{2}$ m

= 0.7 m

As we know, the circumference of the circle = 2πr

$2 \times \dfrac{22}{7} \times 0.7$

$= \dfrac{44}{7} \times 0.7$

$= \dfrac{30.8}{7}$

$= 4.4 \text{ m}$

Hence, the circumference of the roller is 4.4 m.

(ii) Let the roller make a revolutions while traveling 61.6 m.

Distance travelled by roller in a revolutions = Circumference of the wheel x a

⇒ 61.6 = 4.4 x a

⇒ a = $\dfrac{61.6}{4.4}$

⇒ a = 14

Hence, the roller makes 14 revolutions while traveling 61.6 m.

Question 15

Find the area of the circle, length of whose circumference is equal to the sum of the lengths of the circumferences with radii 15 cm and 13 cm.

Answer

Given:

Length of circumference = Sum of the lengths of the circumferences with radii 15 cm and 13 cm

Let `r` be the radius of the circle.

As we know, the circumference of the circle = 2πr

So,

⇒ 2πr = 2π x 15 + 2π x 13

$$\Rightarrow 2 \times \dfrac{22}{7} \times r = 2 \times \dfrac{22}{7} \times 15 + 2 \times \dfrac{22}{7} \times 13$$

$$\Rightarrow \dfrac{44}{7} \times r = \dfrac{44}{7} \times 15 + \dfrac{44}{7} \times 13$$

$$\Rightarrow \dfrac{44}{7} \times r = \dfrac{660}{7} + \dfrac{572}{7}$$

$$\Rightarrow \dfrac{44}{7} \times r = \dfrac{660 + 572}{7}$$

$$\Rightarrow \dfrac{44}{7} \times r = \dfrac{1,232}{7}$$

$$\Rightarrow r = \dfrac{1,232 \times 7}{7 \times 44}$$

$$\Rightarrow r = \dfrac{1,232 \times \cancel{7}}{\cancel{7} \times 44}$$

$$\Rightarrow r = \dfrac{1,232}{44}$$

$$\Rightarrow r = 28 \text{ cm}$$

So, radius of the circle = 28 cm

And, area of the circle = πr^2

$$= \frac{22}{7} \times 28^2$$

$$= \frac{22}{7} \times 784$$

$$= \frac{17,248}{7}$$

$$= 2,464 \text{ cm}^2$$

Hence, the area of the circle is 2,464 cm^2.

Question 16

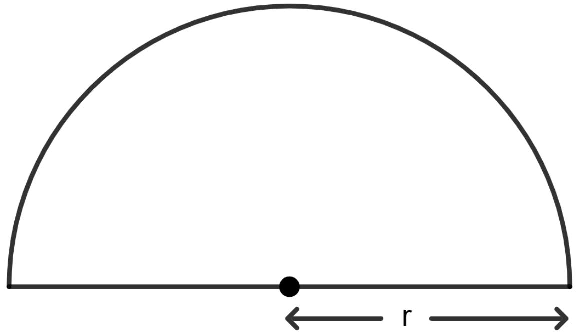
A piece of wire of length 108 cm is bent to form a semicircular arc bounded by its diameter. Find its radius and area enclosed.

Answer

Given:

Total length of the wire = 108 cm

Let `r` be the radius of the circle.



Total length of wire = Circumference of the semicircle + diameter

As we know, the circumference of the semicircle = πr

$$\Rightarrow 108 = \frac{22}{7} \times r + 2r$$

$$\Rightarrow 108 = \frac{22}{7}r + \frac{2}{1}r$$

$$\Rightarrow 108 = \frac{22}{7}r + \frac{2 \times 7}{1 \times 7}r$$

$$\Rightarrow 108 = \frac{22}{7}r + \frac{14}{7}r$$

$$\Rightarrow 108 = \frac{22 + 14}{7}r$$

$$\Rightarrow 108 = \frac{36}{7}r$$

$$\Rightarrow r = \frac{7 \times 108}{36}$$

$\Rightarrow r = \dfrac{756}{36}$

$\Rightarrow r = 21 \text{ cm}$

And, area of the semicircle = $\dfrac{1}{2}\pi r^2$

$= \dfrac{1}{2} \times \dfrac{22}{7} \times 21^2$

$= \dfrac{22}{14} \times 441$

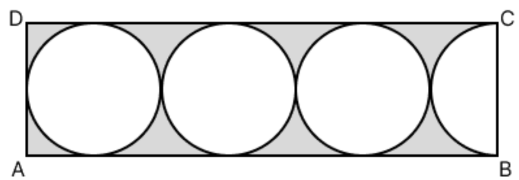
$= \dfrac{9,702}{14}$

$= 693 \text{ cm}^2$

Hence, the radius of the circle is 21 cm and the area is 693 cm^2.

Question 17

In the following figure, a rectangle ABCD encloses three circles. If BC = 14 cm, find the area of the shaded portion. $(\text{Take } \pi = 3\frac{1}{7})$



Answer

Given:

BC = 14 cm

AB = 14 + 14 + 14 + 7

= 49 cm

Area of shaded portion = Area of rectangle ABCD - (3 x Area of circle + Area of semicircle)

As we know, the area of rectangle = length x breadth

= 49 x 14 cm^2

= 686 cm^2

Diameter of the circle = 14 cm

Radius of the circle = $\frac{14}{2}$ cm = 7 cm

Area of the circle = πr^2

$$\text{Area of the circle} = \frac{22}{7} \times 7^2$$

$$= \frac{22}{7} \times 49$$

$$= \frac{1078}{7}$$

$$= 154 \text{ cm}^2$$

Area of 3 circles = 3 x 154 cm^2 = 462 cm^2

Area of the semicircle = $\frac{1}{2}\pi r^2$

$$= \frac{1}{2} \times \frac{22}{7} \times 7^2$$

$$= \frac{22}{14} \times 49$$

$$= \frac{1078}{14}$$

$$= 77 \text{ cm}^2$$

⇒ Area of shaded portion = 686 - (462 + 77) cm^2

= 686 - 539 cm^2

= 147 cm^2

Hence, the area of the shaded portion is 147 cm^2.

Test Yourself

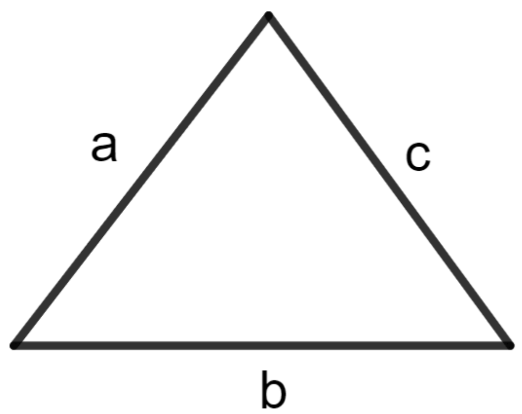
Question 1(i)

Each side of a triangle is doubled, the ratio of areas of the original triangle to the new (resulting) triangle is :

1. 1 : 1
2. 4 : 1
3. 1 : 4
4. 1 : 2

Answer

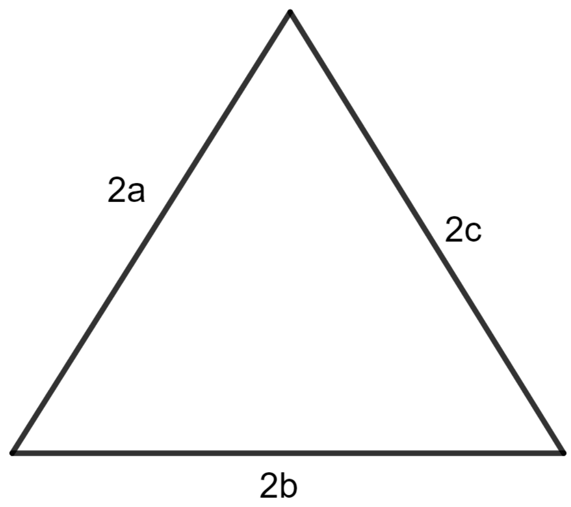
Let a, b and c be sides of the triangle.



$$s = \frac{a+b+c}{2}$$

Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

When each side of a triangle is doubled, then 2a, 2b and 2c are sides of new triangle.



$$s' = \frac{a+b+c}{2}$$

$$s' = \frac{2a+2b+2c}{2}$$

$$s' = \frac{2(a+b+c)}{2}$$

$$s' = 2\frac{(a+b+c)}{2}$$

$$s' = 2s$$

Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

= $\sqrt{2s(2s-2a)(2s-2b)(2s-2c)}$

= $\sqrt{2s2(s-a)2(s-b)2(s-c)}$

= $\sqrt{16s(s-a)(s-b)(s-c)}$

= 4 $\sqrt{s(s-a)(s-b)(s-c)}$

Ratio of original triangle to new triangle = $\dfrac{\sqrt{s(s-a)(s-b)(s-c)}}{4\sqrt{s(s-a)(s-b)(s-c)}}$

= $\dfrac{1}{4}$

Hence, option 3 is the correct option.

Question 1(ii)

The perimeter of a rectangle is equal to the perimeter of a square. If each side of the square is 30 cm and length of the rectangle is 40 cm; the breadth of the rectangle is :

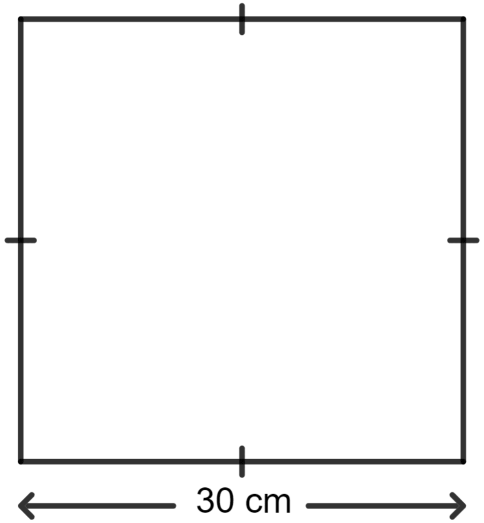
1. 5 cm
2. 16 cm
3. 15 cm
4. 20 cm

Answer

Given:

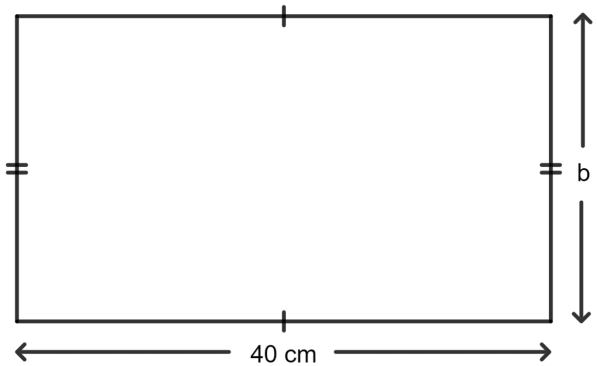
The perimeter of a rectangle = The perimeter of a square

Each side of the square = 30 cm



Length of the rectangle = 40 cm

Let b be the breadth of the rectangle.



As we know that perimeter of the rectangle = 2(length + breadth)

And, perimeter of the square = 4 x side

⇒ 2(40 + b) = 4 x 30

⇒ 80 + 2b = 120

⇒ 2b = 120 - 80

⇒ 2b = 40

⇒ b = $\dfrac{40}{2}$

⇒ b = 20

Hence, option 4 is the correct option.

Question 1(iii)

The area of a trapezium is 80 square unit. If its two parallel sides are 6 unit and 14 unit; the distance between parallel sides is :

1. 4 unit
2. 8 unit
3. 12 unit
4. 1 unit

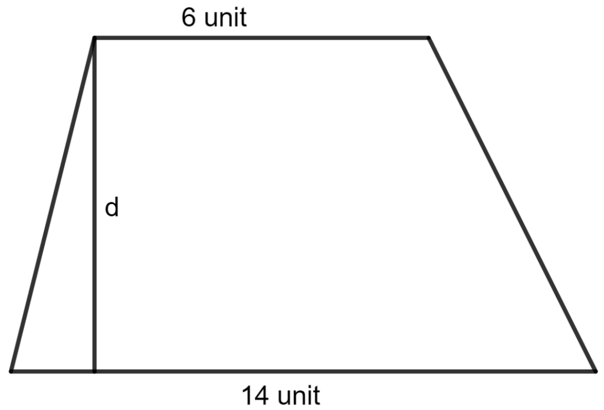
Answer

Given:

The lengths of the parallel sides of the trapezium are 6 unit and 14 unit.

The area of a trapezium = 80 square unit

Let d be the distance between parallel sides.



As we know, the area of a trapezium = $\frac{1}{2}$ x (sum of parallel sides) x height

⇒ $\frac{1}{2}$ x (6 + 14) x d = 80

⇒ $\frac{1}{2}$ x 20 x d = 80

⇒ 10 x d = 80

⇒ d = $\frac{80}{10}$

⇒ d = 8 unit

Hence, option 2 is the correct option.

Question 1(iv)

The external and internal radii of a ring shaped metal sheet are 10 cm and 8 cm respectively. The area of its one face, in terms of π is :

1. 36π cm^2

2. 64π cm^2

3. 100π cm^2

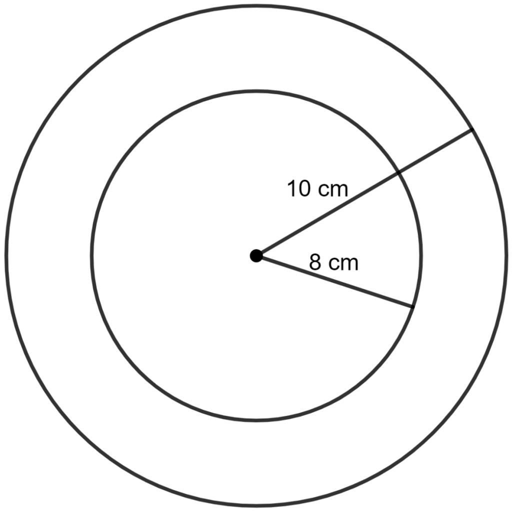
4. 168π cm^2

Answer

Given:

Radius of outer circle = 10 cm

Radius of inner circle = 8 cm



Area of circular ring = Area of outer circle - Area of inner circle

As we know, the area of the circle = πr^2

⇒ Area of circular ring = $\pi \times (10)^2 - \pi \times (8)^2$

= 100π - 64π

= 36π cm^2

Hence, option 1 is the correct option.

Question 1(v)

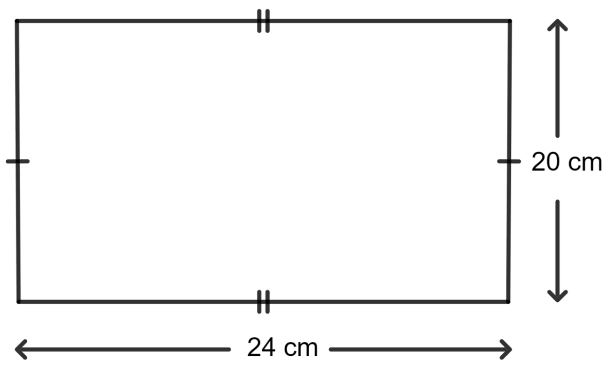
The adjacent sides of a rectangle are 20 cm and 24 cm. If its perimeter is equal to the circumference of a circle, the radius of the circle is :

1. 14 cm
2. 21 cm
3. 28 cm
4. 7 cm

Answer

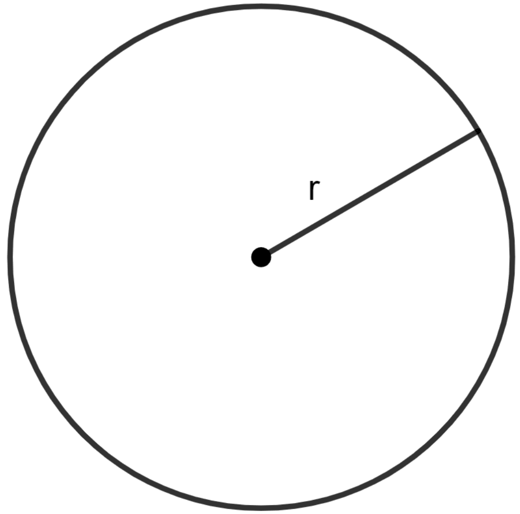
Given:

The adjacent sides of a rectangle are 20 cm and 24 cm.



The perimeter of rectangle = The circumference of a circle

Let r be the radius of the circle.



As we know, the perimeter of the rectangle = 2(length + breadth)

And, the circumference of the circle = 2πr

⇒ 2(20 + 24) = 2 x $\dfrac{22}{7}$ x r

⇒ 2 x 44 = $\dfrac{44}{7}$ x r

⇒ 88 = $\dfrac{44}{7}$ x r

⇒ r = $\dfrac{7 \times 88}{44}$

⇒ r = $\dfrac{616}{44}$

⇒ r = 14 cm

Hence, option 1 is the correct option.

Question 1(vi)

Statement 1: ABCD is a parallelogram, base AB = 12 cm, perpendicular dropped from D to AB is 9 cm, area of parallelogram ABCD = 54 cm^2.

Statement 2: The area of parallelogram = Base x Height.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

Given,

AB (base) = 12 cm, perpendicular dropped from D to AB (height) = 9 cm.

By formula,

The area of parallelogram = Base x Height

= 12 x 9

= 108 cm^2.

∴ Statement 1 is false, and statement 2 is true.

Hence, option 4 is the correct option.

Question 1(vii)

Assertion (A) : The diagonal of a rectangle is 17 m and its breadth is 8 m. Half of the area of rectangle is 120 m^2.

Reason (R) : The diagonal of every rectangle, divide into two congruent right-triangles.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

The diagonal of the rectangle = 17 m

The breadth of the rectangle = 8 m

Using Pythagoras theorem in rectangle,

⇒ $\text{Diagonal}^2 = \text{Length}^2 + \text{Breadth}^2$

⇒ $17^2 = l^2 + 8^2$

⇒ $289 = l^2 + 64$

⇒ $l^2 = 289 - 64$

⇒ $l^2 = 225$

⇒ $l = \sqrt{225}$

⇒ $l = 15$ m.

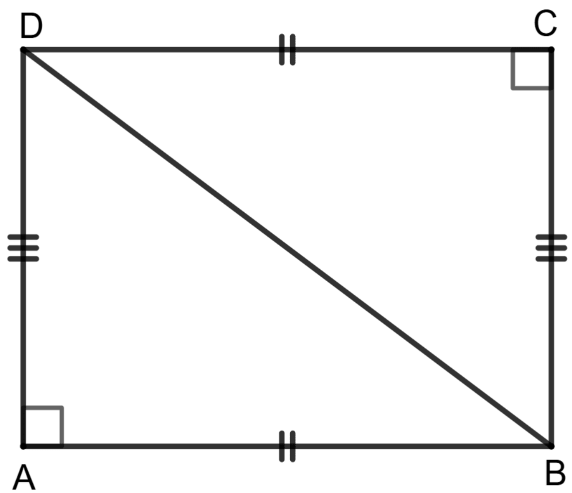
By formula,

Area of rectangle = l x b = 15 x 8 = 120 m^2.

Half of the area of rectangle = $\dfrac{120}{2}$ = 60 m^2.

So, assertion (A) is false.

ABCD is a rectangle. BD is a diagonal.



In Δ ABD and Δ BDC,

⇒ AB = DC (Opposite sides of rectangle are equal)

⇒ AD = BC (Opposite sides of rectangle are equal)

⇒ BD = BD (Common)

∴ Δ ABD ≅ Δ BDC (By SSS congruency criterion)

Thus the diagonal of every rectangle, divide into two congruent right-triangles.

So, reason (R) is true.

∴ A is false, but R is true.

Hence, option 4 is the correct option.

Question 1(viii)

Assertion (A) : The area of an equilateral triangle of height $2\sqrt{3}$ is $4\sqrt{3}$ cm^2.

Reason (R) : The area of an equilateral triangle = $\dfrac{1}{\sqrt{3}}$ x (Height)^2.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given, height of the equilateral triangle = 2 $\sqrt{3}$ cm

By formula,

The area of an equilateral triangle = $\dfrac{1}{\sqrt{3}}$ x (Height)^2

So, reason (R) is true.

Substituting values we get :

$$\text{Area} = \frac{1}{\sqrt{3}} \times (2\sqrt{3})^2$$

$$= \frac{1}{\sqrt{3}} \times 4 \times 3$$

$$= 4\sqrt{3} \text{ cm}^2.$$

So, assertion (A) is true.

∴ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 1(ix)

Assertion (A) : The diagonal of a square is $9\sqrt{2}$ cm. Its area = 81 cm^2.

Reason (R) : The area of a square = $\dfrac{1}{2}$ x d^2, where d is the length of the diagonal

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given, the diagonal of a square = $9\sqrt{2}$ cm

By formula,

The area of a square = $\dfrac{1}{2}$ x d^2, where d is the length of the diagonal.

So, reason (R) is true.

Given,

The diagonal of a square is $9\sqrt{2}$ cm.

$$\text{Area of square } = \dfrac{1}{2} \times (9\sqrt{2})^2$$

$$= \dfrac{1}{2} \times 81 \times 2$$

$$= 81 \text{ cm}^2.$$

So, assertion (A) is true.

∴ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 1(x)

Assertion (A) : The area of trapezium with base 10 cm, height 5 cm and the side parallel to the given base being 6 cm is 40 cm^2.

Reason (R) : The area of trapezium = $\dfrac{1}{2}$ x (sum of non parallel sides) x height.

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are incorrect.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

Given,

Base = 10 cm

Height = 5 cm

The side parallel to the given base = 6 cm

By formula,

Area of trapezium = $\dfrac{1}{2}$ x (sum of parallel sides) x height.

$\text{Area of given trapezium} = \dfrac{1}{2} \times (10 + 6) \times 5$

$= \dfrac{1}{2} \times 16 \times 5$

$= 8 \times 5$

$= 40 \text{ cm}^2.$

∴ A is true, but R is false.

Hence, option 3 is the correct option.

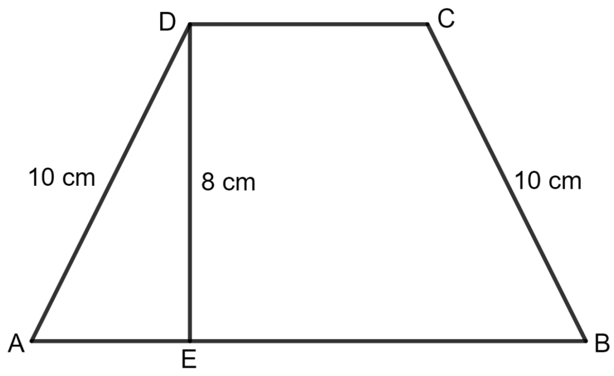
Question 2

The perimeter of a trapezium is 52 cm. If its non-parallel sides are 10 cm each and its altitude is 8 cm, find the area of the trapezium.

Answer

The perimeter of a trapezium = 52 cm.

The altitude = 8 cm



Perimeter of the trapezium = AB + BC + CD + DA

⇒ 52 = AB + 10 + CD + 10

⇒ 52 = AB + CD + 20

⇒ AB + CD = 52 - 20

⇒ AB + CD = 32

⇒ Sum of parallel sides = 32

As we know, the area of a trapezium = $\frac{1}{2}$ x (sum of parallel sides) x height

= $\frac{1}{2}$ x 32 x 8 cm^2

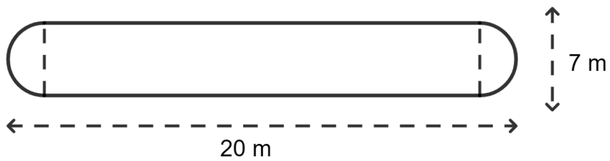
= $\frac{1}{2}$ x 256 cm^2

= 128 cm^2

Hence, the area of the trapezium is 128 cm^2.

Question 3

The shape of a garden is rectangular in the middle and semicircular at each end as shown in the figure. Find the area and perimeter of the garden.



Answer

Given:

Total length of garden = 20 m

Radius of the circle = $\frac{7}{2}$ m

Length of rectangle = total length - 2 x radius of the circle

= 20 - 2 x $\frac{7}{2}$ m

= 20 - 7 m

= 13 m

Perimeter of the figure = l + l + 2 x $\frac{1}{2}$ x circumference of semicircle

As we know that circumference of the circle = 2πr

= 13 + 13 + 2 x $\frac{1}{2}$ x 2πr

= 26 + 2 x $\frac{22}{7} \times \frac{7}{2}$

= 26 + $\cancel{2} \times \frac{22}{\cancel{7}} \times \frac{\cancel{7}}{\cancel{2}}$

= 26 + 22

= 48 m

Area of the figure = Area of rectangle + 2 x Area of semicircle

As we know that area of rectangle = length x breadth

And, area of the circle = πr^2

So, area

$$= 13 \times 7 + \frac{22}{7} \times (\frac{7}{2})^2$$

$$= 91 + \frac{22}{7} \times \frac{49}{4}$$

$$= 91 + \frac{22 \times 49}{7 \times 4}$$

$$= 91 + \frac{1078}{28}$$

$= 91 + 38.5$

$= 129.5 \text{ m}^2$

Hence, the area of the garden is 129.5 m^2 and the perimeter is 48 m.

Question 4

Each side of rhombus is 13 cm and one of its diagonals is 10 cm. Find :

(i) the length of its other diagonal

(ii) its area.

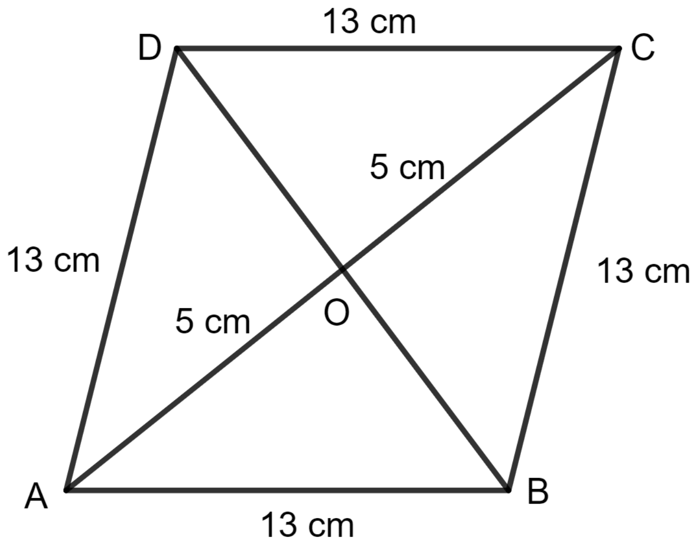
Answer

(i) **Given :**

AB = 13 cm

AC = 10 cm

Then, OA = OC = $\dfrac{10}{2}$ = 5 cm



Since the diagonals of a rhombus bisect at 90°.

Applying pythagoras theorem in triangle AOB, we get:

$AB^2 = OA^2 + OB^2$

$\Rightarrow (13)^2 = (5)^2 + OB^2$

$\Rightarrow 169 = 25 + OB^2$

$\Rightarrow OB^2 = 169 - 25$

$\Rightarrow OB^2 = 144$

$\Rightarrow OB = \sqrt{144}$

$\Rightarrow OB = 12$

BD = 2 x OB

= 2 x 12 cm

= 24 cm

Hence, the length of other diagonal is 24 cm.

(ii) As we know, the area of rhombus = $\frac{1}{2}$ x product of its diagonal

= $\frac{1}{2}$ x 10 x 24 cm^2

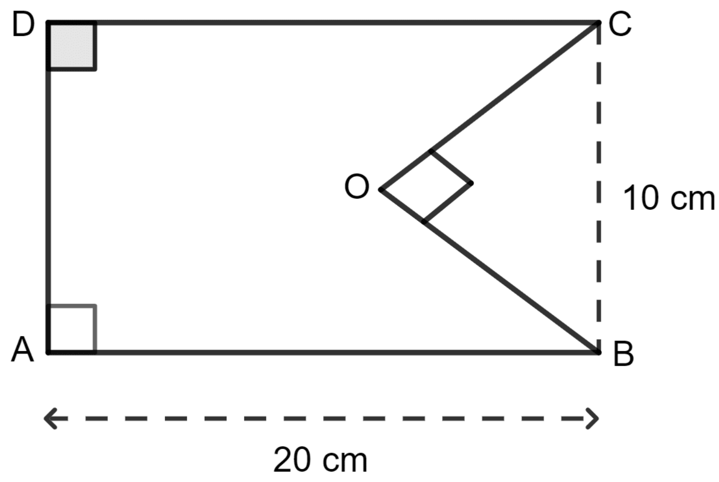
= $\frac{1}{2}$ x 240 cm^2

= 120 cm^2

Hence, the area of the rhombus is 120 cm^2.

Question 5

The given figure shows a rectangle ABCD and a right triangle BOC. Find the area of the shaded portion.



Answer

Given:

Length of rectangle = 20 cm

Breadth of rectangle = 10 cm

Height of triangle = 8 cm

Hypotenuse of triangle = 10 cm

Applying pythagoras theorem in triangle COB, we get:

$BC^2 = OB^2 + OC^2$

$\Rightarrow (10)^2 = OB^2 + (8)^2$

$\Rightarrow 100 = OB^2 + 64$

$\Rightarrow OB^2 = 100 - 64$

⇒ OB^2 = 36

⇒ OB = $\sqrt{36}$

⇒ OB = 6 cm

Area of shaded figure = area of rectangle - area of triangle

As we know that area of rectangle = length x breadth

And, area of the triangle = $\frac{1}{2}$ x base x height

So, Area

$$= 20 \times 10 - \frac{1}{2} \times 6 \times 8$$

$$= 200 - \frac{1}{2} \times 48$$

$$= 200 - 24$$

$$= 176 \text{ cm}^2$$

Hence, the area of the shaded portion is 176 cm^2.

Question 6(i)

Find the area of the circle whose circumference is 264 cm.

Answer

Let `r` be the radius of the circle.

The circumference of a circle = 264 cm

As we know, the circumference of the circle = 2πr

$$\Rightarrow 2 \times \frac{22}{7} \times r = 264$$

$$\Rightarrow \frac{44}{7} \times r = 264$$

$$\Rightarrow r = \frac{264 \times 7}{44}$$

$$\Rightarrow r = \frac{1848}{44}$$

$$\Rightarrow r = 42 \text{ cm}$$

And, area of the circle = πr^2

$$= \frac{22}{7} \times 42^2$$

$$= \frac{22}{7} \times 1764$$

$$= \frac{38808}{7}$$

$$= 5544 \text{ cm}^2$$

Hence, the area of the circle is 5,544 cm^2.

Question 6(ii)

Find the circumference of a circle whose area is 1386 cm^2.

Answer

Let `r` be the radius of the circle.

The area of circle = 1386 cm^2

As we know, the area of the circle = πr^2

$$\Rightarrow \frac{22}{7} \times r^2 = 1386$$

$$\Rightarrow r^2 = \frac{1386 \times 7}{22}$$

$$\Rightarrow r^2 = \frac{9702}{22}$$

$$\Rightarrow r^2 = 441$$

$$\Rightarrow r = \sqrt{441}$$

$$\Rightarrow r = 21 \text{ cm}$$

Circumference of the circle = $2\pi r$

$$= 2 \times \frac{22}{7} \times 21$$

$$= \frac{44}{7} \times 21$$

$$= \frac{924}{7}$$

$$= 132 \text{ cm}$$

Hence, the circumference of the circle is 132 cm.

Question 7

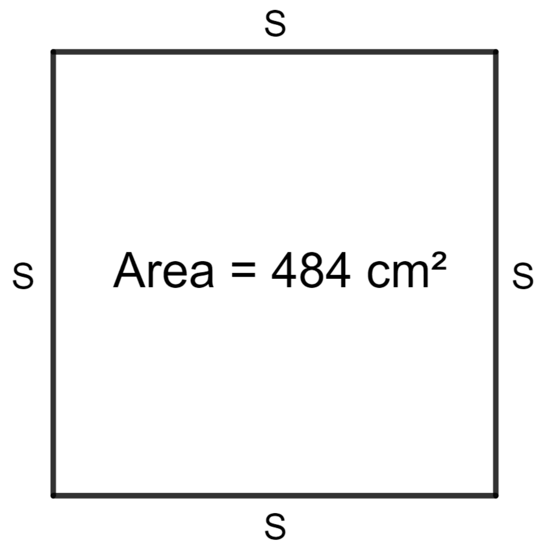
A copper wire when bent in the form of a square encloses an area of 484 cm^2. When the same wire is bent in the form of a circle, find the area of the circle formed.

Answer

Given:

Area of the square = 484 cm^2

Let s be the side of the square.



As we know, the area of the square = $side^2$

⇒ $s^2 = 484$

⇒ $s = \sqrt{484}$

⇒ s = 22 cm

Total length of the wire = Perimeter of the square

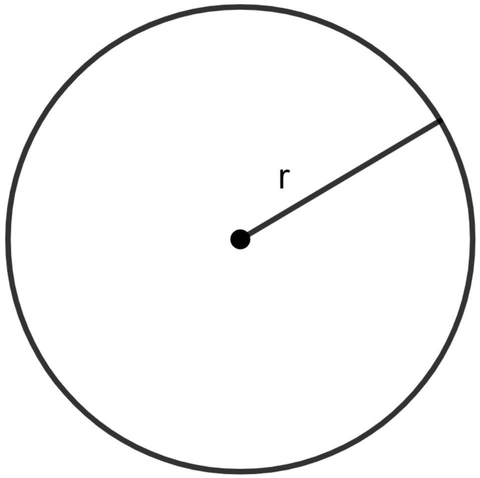
As we know, the perimeter of the square = 4 x side

= 4 x 22

= 88 cm

Perimeter of the square = Circumference of the circle

Let r be the radius of the circle.



⇒ 2πr = 88 cm

$$\Rightarrow 2 \times \frac{22}{7} \times r = 88$$

$$\Rightarrow \frac{44}{7} \times r = 88$$

$$\Rightarrow r = \frac{7 \times 88}{44}$$

$$\Rightarrow r = \frac{616}{44}$$

$$\Rightarrow r = 14 \text{ cm}$$

Area of the circle = πr^2

$$= \frac{22}{7} \times 14^2$$

$$= \frac{22}{7} \times 196$$

$$= \frac{4312}{7}$$

$$= 616 \text{ cm}^2$$

Hence, the area of the circle is 616 cm^2.

Question 8

The perimeters of two squares are 120 cm and 64 cm. Find the perimeter of the square whose area is equal to the sum of the areas of these two squares.

Answer

Given:

The perimeters of two squares are 120 cm and 64 cm.

Let a be the side of the square.

As we know that perimeter of the square = 4 x side

⇒ 4 x a = 120

⇒ a = $\frac{120}{4}$

⇒ a = 30 cm

The perimeter of square is 64 cm.

Let `b` be the side of the square.

As we know that perimeter of the square = 4 x side

⇒ 4 x b = 64

⇒ a = $\frac{64}{4}$

⇒ a = 16 cm

It is given that area of third square = the sum of the areas of these two squares

Let `R` be the side of the square.

As we know that area of the square = side^2

⇒ $R^2 = 30^2 + 16^2$

⇒ $R^2 = 900 + 256$

⇒ $R^2 = 1156$

⇒ R = $\sqrt{1156}$

⇒ R = 34 cm

Perimeter of third square = 4 x 34 cm

= 136 cm

Hence, the perimeter of the square is 136 cm.

Question 9

Find the perimeter of a square whose area is equal to half the sum of areas of three squares with sides 3 cm, 4 cm and 5 cm.

Answer

Given:

Area of square = Half the sum of areas of three squares with sides 3 cm, 4 cm and 5 cm.

Let `R` be the side of the square.

As we know that area of the square = side^2

$$\Rightarrow R^2 = \frac{1}{2} \times (3^2 + 4^2 + 5^2)$$

$$\Rightarrow R^2 = \frac{1}{2} \times (9 + 16 + 25)$$

$$\Rightarrow R^2 = \frac{1}{2} \times 50$$

$$\Rightarrow R^2 = 25$$

$\Rightarrow R = \sqrt{25}$

$\Rightarrow R = 5 \text{ cm}$

And, perimeter of the square = 4 x side

Perimeter of the square = 4 x 5 cm

= 20 cm

Hence, the perimeter of the square is 20 cm.

Question 10

If the diameter of a car wheel is 28 cm, how many times will the wheel rotate in a journey of 11 km ?

Answer

Given:

Diameter of wheel = 28 cm

Radius = $\dfrac{\text{Diameter}}{2}$

= $\dfrac{28}{2}$ cm

= 14 cm

As we know, the circumference of the circle = 2πr

$2 \times \dfrac{22}{7} \times 14$

$= \dfrac{44}{7} \times 14$

$= \dfrac{616}{7}$

$= 88 \text{ cm}$

Let a be the number of revolutions.

Total distance = 11 km

= 11 x 1,00,000 cm

= 11,00,000 cm

Distance traveled by wheel in a revolutions = circumference of the wheel x a

⇒ 11,00,000 = 88 x a

⇒ a = $\dfrac{11,00,000}{88}$

⇒ a = 12,500 revolutions

Hence, the wheel rotates 12,500 times during a journey of 11 km.

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